A COGNITIVE-TAXONOMY HANDBOOK

Bloom's Taxonomy for Rasashastra & Bhaishajya Kalpana

Topic-by-topic domain and level mapping, objective design and usage guide for AyUG-RB — Second Professional BAMS
Ayurvediya Aushadhi Nirmana & Prayoga Vigyana

Curriculum	NCISM, Board of Ayurveda — applicable from the 2021–22 batch, academic year 2023–24 onwards
Coverage	2 papers · 26 theory topics · 10 practical groups · 268 objectives coded in both taxonomies
Framework	Bloom (1956) and the Anderson & Krathwohl revision (2001) carried in parallel, with the psychomotor and affective domains, read against NCISM Table 3 Column C3 and Table 4 Column C4

268	203	33	32
OBJECTIVES CODED	COGNITIVE	PSYCHOMOTOR	AFFECTIVE

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, objective writing, assessment design and paper setting.

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Unit 8 Rasa Dravya Parichaya- I 9 obj · 10 marks 35

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PART A

Understanding the taxonomy

Eight short chapters. The first four explain what Bloom's taxonomy is, what its two versions are, what its three domains cover, and where it already sits inside the AyUG-RB curriculum. The next three turn it into practice: choosing verbs, choosing how to teach, choosing how to check. The last explains how to read a Unit in Part B. Read straight through once; after that, use the Units and come back here only when something needs settling.

1. **What Bloom's taxonomy is, and why this book carries two versions**
2. **The three domains** — thinking, doing, and how a student goes about the work
3. **The second axis** — what kind of knowledge an objective works on
4. **Where Bloom already sits in your curriculum** — reading Columns C3 and C4
5. **Verbs** — the working end of the taxonomy
6. **Choosing how to teach**, level by level
7. **Choosing how to check**, level by level
8. **How to use a Unit in Part B**

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PART A · CHAPTER 1

What Bloom's taxonomy is, and why this book carries two versions

In 1956 a committee of American examiners led by Benjamin Bloom published a way of sorting examination questions. They had noticed something every teacher notices sooner or later: two questions on the same topic can be worlds apart in difficulty, not because one covers more ground, but because one asks the student to *do* something harder with what they know. "Name the four types" and "decide which of the four applies here, and say why" sit on the same syllabus line. They are not the same question.

What the committee produced was a ladder of six levels, each asking more of the student than the one below: **Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation**. The ladder is cumulative. You cannot apply what you do not understand, and you cannot understand what you cannot recall. That single idea — that the levels stack — is what makes the taxonomy useful for planning rather than merely for labelling.

THE 2001 REVISION

Forty-five years later a group led by Lorin Anderson and David Krathwohl — Krathwohl had worked on the original — published a revision. Three things changed.

- The levels became **verbs** rather than nouns: *Remember, Understand, Apply, Analyse, Evaluate, Create*. A taxonomy of thinking should be named by what the thinker does.
- **The top two levels swapped.** Bloom put Synthesis at level 5 and Evaluation at level 6. The revision reversed them: Evaluate is level 5, Create is level 6. The reasoning was that making something new is a larger act than judging something that already exists.
- A **second axis** was added, for the kind of knowledge being worked on. Chapter 3 covers it.

> **This is the one thing to carry away from this chapter.** When a colleague says "that question is at the top level", ask which scheme they mean. In the 1956 scheme the top level is Evaluation — judging. In the 2001 scheme it is Create — producing. Those are different questions, different teaching and different marking. Most confusion between two well-informed teachers about Bloom comes down to this.

WHY BOTH, IN THIS BOOK

The AyUG-RB curriculum uses the **1956 scheme**. Its codes are CK, CC, CAP, CAN, CS and CE — Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation. If this book used only the 2001 names, you could not match it against your own syllabus document.

But most faculty development programmes now teach the 2001 scheme, most recent writing on assessment uses it, and younger colleagues will have met it first. If this book used only the 1956 names, it would be teaching a vocabulary its readers are being trained out of.

So every level in this book is named twice, in a fixed order: the 1956 name first, because that is what your curriculum prints, then the 2001 name. *Comprehension / Understand. Evaluation / Evaluate.* Where the two schemes disagree about position — only at the top two levels — the book says so on the page rather than quietly picking one.

PART A · CHAPTER 2

The three domains — thinking, doing, and how a student goes about the work

Bloom's six levels describe thinking only. The original committee knew that thinking is not all of learning, and planned three volumes. The **cognitive** volume — thinking — appeared in 1956. The **affective** volume — attitudes, values, how a student goes about the work — appeared in 1964, edited by Krathwohl. The **psychomotor** volume — physical skill — was never written by that committee at all; others filled the gap later, and the version your curriculum uses comes from that later work.

All three matter in Rasashastra and Bhaishajya Kalpana, and a pharmacy subject probably needs all three more than most. A student who can recite a procedure flawlessly, cannot carry it out, and does not much care whether the material is genuine has met one domain out of three.

THINKING — THE COGNITIVE DOMAIN, SIX LEVELS

Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation, as Chapter 1 set out. This is the domain the theory papers examine, and it accounts for most of the objectives in the curriculum.

DOING — THE PSYCHOMOTOR DOMAIN, FIVE LEVELS AS USED HERE

Physical skill has its own ladder, and it does not map onto the thinking one. Your curriculum uses five levels:

- **Set** — ready to begin. Knows what is needed and can lay it out.
- **Guided response** — performs while being shown or guided.
- **Mechanism** — performs reliably alone, without prompting.
- **Adaptation** — changes the method when conditions change.
- **Origination** — devises a procedure that was not given.

The published psychomotor schemes have more levels than five; these are the five the curriculum uses. Note where the real work sits: getting a student from *Guided response* to *Mechanism* is most of what practical hours are for, and it is slow. Demonstrating well takes one session. Withdrawing your help gradually until the student is reliable alone takes many.

GOING ABOUT IT — THE AFFECTIVE DOMAIN, FIVE LEVELS

- **Receiving** — pays attention when it matters.
- **Responding** — joins in willingly.
- **Valuing** — treats it as worth doing, unprompted.
- **Organisation** — fits it together with other things they value.
- **Characterisation** — it has become how they work.

> **The affective domain needs a different kind of evidence, and this is where it is usually lost.** You cannot see an attitude in a single answer on a single afternoon. Valuing shows up as what a student does when nobody is checking — whether they verify the material, whether they record the reading they actually took. That evidence is gathered by watching over time: conduct in the practical, a short reflection, a checklist applied more than once across a term. An examination hall cannot produce it. So affective objectives have to be planned for in advance, or they go unassessed while appearing to have been assessed.

THE DOMAINS DO NOT LINE UP

An objective at Mechanism — performing reliably alone — may need only level 1 thinking, if the procedure is followed by rote. Another at Set — merely getting ready — may need level 4 thinking, if choosing the right equipment requires judgement. Do not assume the three ladders move together. The Units in Part B show all three separately for exactly this reason.

PART A · CHAPTER 3

The second axis — what kind of knowledge an objective works on

The 2001 revision added a question the 1956 scheme did not ask. The first question is: what thinking must the student do? The second is: *what kind of knowledge is it about?* Cross the two and you get a grid rather than a ladder, and the grid explains something teachers meet constantly — that two objectives at the same level can need completely different teaching.

THE FOUR KINDS

- **Factual** — terminology, names, lists, specific details. The vocabulary of the subject. Learned by rehearsal and self-testing, and by very little else.
- **Conceptual** — classifications, principles, relationships, the reason a thing is so. Learned by explaining it to somebody, which is why a student who has only read it can rarely produce it.
- **Procedural** — methods, techniques, criteria for deciding when to use which. Learned by carrying it out. Reading a procedure is not learning it.
- **Metacognitive** — knowing about your own learning: what you find hard, which approach works for you, when to change tack. Learned by being made to look back at how you worked.

WHY THIS CHANGES THE TEACHING

Take two objectives, both at level 1. One asks a student to recall the names in a group; the other asks them to recall the steps of a procedure in order. Same level, and the first is Factual while the second is Procedural. Rehearsal will fix the first. The second will not hold until the student has done it, because a remembered list of steps and a performable procedure are not the same possession.

This is also why "they know it but they cannot do it" is such a common complaint in practical subjects. Often the objective was Procedural and the teaching was pitched as though it were Factual.

> **How the Units read this.** Each objective in Part B is labelled with its kind of knowledge, worked out from the words of the objective itself — its verb, and what the verb acts on. Where an objective genuinely works on two kinds at once, both are named: "Factual + Procedural" means part of it is learned by rehearsal and part by doing. Where the wording supports a reading but does not settle it, the label carries a small circle (°), so you can see that a judgement was made and overrule it if you disagree. Nothing is guessed silently.

ONE THING WORTH NOTICING

The Metacognitive row is empty across the whole of AyUG-RB, as is the Create level. That is simply what the curriculum asks for at this stage, and no individual teacher needs to fix it. It is recorded here because it is the kind of thing worth having in front of you when objectives are next revised, and because a teacher who wants to build reflective habit in students should know they are going beyond what the syllabus requires rather than filling a gap in it.

PART A · CHAPTER 4

Where Bloom already sits in your curriculum

Bloom is not something this book adds to AyUG-RB. It is already there, and has been since the curriculum was published. Most teachers have simply never had it pointed out.

COLUMN C3 AND COLUMN C4

Open the curriculum at Table 3, the theory objectives. The third column is headed *Domain/sub*. Every objective carries a code there: CK, CC, CAP, CAN, CS, CE for thinking; PSY-SET, PSY-GUD, PSY-MEC, PSY-ADT, PSY-ORG for doing; AFT-REC, AFT-RES, AFT-VAL, AFT-SET, AFT-CHR for how the student goes about it. Table 4, the practical objectives, carries the same codes in Column C4.

Those sixteen codes are Bloom, the psychomotor scheme, and Krathwohl's affective scheme. The full key is in Appendix C2. What matters here is that the curriculum has already decided, for each of its 268 objectives, what kind of learning it is asking for — and that this decision is available to you before you plan a single session.

WHAT THE OTHER COLUMNS GIVE YOU

- **Column D3 / D4 — Must know, Desirable to know, Nice to know.** An independent judgement about priority. It is not part of Bloom and does not move with level: an objective can be Must know at level 1 and Nice to know at level 6, or the reverse.
- **Column F3 / F4 — teaching and learning method.** The curriculum names methods for each objective. Chapter 6 sets these against the levels.
- **Column G3 / G4 — assessment.** The tools named for each objective. Chapter 7 sets these against the levels.
- **Column I3 / I4 — term.** When in the year the objective falls.

> **A word about Column E3, which this book does not use.** Column E3 is also headed *Level*, but it carries a different scheme — Knows, Knows how, Shows how, Does — which is Miller's pyramid, not Bloom. It answers a different question: not what kind of thinking is needed, but how authentic the evidence has to be. It is worth knowing well, and it has its own companion volume. In *this* book, "level" always means the Bloom level given in Column C3 or C4. Where you see "level 5" here, it never means "Shows how".

READING A CODE IN PRACTICE

Suppose an objective carries **CC**, **MK**, and a teaching method of group discussion. Read that as: this is level 2 thinking, Comprehension in the 1956 scheme and Understand in the 2001 one; it is Must know, so no student may leave it behind; and the curriculum expects students to handle the material themselves rather than receive it again. That is already most of a lesson plan, and it came out of three columns.

PART A · CHAPTER 5

Verbs — the working end of the taxonomy

A level is an idea. A verb is what a student actually meets. This is why the 2001 revision renamed the levels as verbs, and it is the most practical thing in the whole taxonomy: **an objective is taught and examined at the level of its verb**, whatever else is written beside it.

Consider two objectives on identical content. *Enlist the types. Justify the choice between the types.* The first is level 1: a student meets it by rehearsal, and answers it in a line. The second is level 6: a student meets it by making a judgement and defending it, and answers it in a page. A student who prepared for the first and met the second in the examination has not been unlucky. They read the objective and it did not tell them what they needed to know.

A WORKING LIST

- **Level 1, Knowledge / Remember** — define, enlist, list, name, recall, recite, state, identify, recognise, label.
- **Level 2, Comprehension / Understand** — describe, explain, discuss, illustrate, classify, summarise, differentiate, distinguish, elaborate, outline.
- **Level 3, Application / Apply** — apply, demonstrate, use, perform, prepare, calculate, solve, employ, conduct, measure.
- **Level 4, Analysis / Analyse** — analyse, compare, contrast, examine, determine, categorise, correlate, differentiate between, select, review.
- **Level 5 in 1956, Synthesis** — design, formulate, develop, construct, compose, devise, plan, produce. (In the 2001 scheme these belong to level 6, Create.)
- **Level 6 in 1956, Evaluation** — evaluate, justify, appraise, critique, assess, judge, defend, validate, interpret. (In the 2001 scheme, level 5.)

The affective domain has its own verbs, and they are not thinking verbs: *appreciate, value, accept, comply, respond, defend, advocate, adhere, display, practise.* If an affective objective opens with *explain*, it will be taught and marked as level 2 thinking, whatever its code says, because *explain* is what the student will be asked to do.

> **Where a verb and a code point in different directions, follow the verb — and know that you are choosing.** The verb is what appears on the question paper, so in practice it wins by default. That is fine, provided it is a decision rather than an accident. If the verb asks for less than the code, you have room to teach higher than the syllabus requires. If it asks for more, plan for the verb, or your students will meet a question they were never prepared for. The Units in Part B tell you, objective by objective, which case you are in.

WHEN YOU WRITE YOUR OWN OBJECTIVES

Pick the verb last and pick it deliberately. Decide the level first, then choose a verb from that level, then check that the verb can actually be assessed by the tool you have in mind. An objective that says *appreciate* and is assessed by a written paper has a mismatch built into it before the term begins. One verb, one level, one objective: where you find yourself writing “define, recite and interpret” in a single line, you have three objectives at three levels, and they will need three different questions.

PART A · CHAPTER 6

Choosing how to teach, level by level

The commonest planning error in a crowded syllabus is to choose the method by the hour rather than by the level — a lecture because there is a lecture slot. Lectures build level 1 well and level 2 poorly, and above level 2 they build almost nothing on their own. What follows is what each level needs.

LEVEL 1 — KNOWLEDGE / REMEMBER

Exposure, then rehearsal, then retrieval practice. A lecture puts the facts in front of students; only repeated recall makes them stay. The strongest tool is the cheapest: make students retrieve without looking, then check. Quizzes, word puzzles and game-based methods — all named in this curriculum — are retrieval practice in a form students will actually do. Spread it across weeks; a single long session before the examination produces recall that lasts until the examination.

LEVEL 2 — COMPREHENSION / UNDERSTAND

Students must handle the material rather than receive it again. Have them sort it, compare it, put it in an order of their own, and explain it to a classmate. Discussion and flipped classroom do this work. Repeating the lecture more slowly does not: the difficulty is not that they did not hear it.

LEVEL 3 — APPLICATION / APPLY

Work examples through together, then hand over a situation the student has not seen. The hand-over is the teaching. Without it students learn the example rather than the method, and it is easy to mistake one for the other, because a class that can complete the worked example looks exactly like a class that has understood.

LEVEL 4 — ANALYSIS / ANALYSE

Give students two things and ask how they differ and why. Analysis needs something to set against something else; a single object cannot be analysed. Where the curriculum already distinguishes sub-categories, the material for this is sitting there.

LEVEL 5 IN 1956, SYNTHESIS — CREATE IN 2001

Not asked for anywhere in this curriculum. If you choose to reach for it, it means asking students to assemble something that was not handed to them, and marking it against criteria set out in advance.

LEVEL 6 — EVALUATION / EVALUATE

Give students something to judge, and tell them what to judge it against. Then require the criteria to be stated, not just the conclusion. Small-group debate works well because a student has to defend a position and hear it challenged, which is the whole of the skill. This is the level most often taught as though it were level 2 — explaining the accepted view rather than reaching a view.

> **The practical hours belong mostly to one level.** In the practical Units, the bulk of the time should go to moving students from *Guided response* to *Mechanism* — from performing with your help to performing reliably without it. That transition is invisible if you only watch the best student in the group, because they arrive at it early. Watch the middle of the group to know where the class actually is.

PART A · CHAPTER 7

Choosing how to check, level by level

A check is fair when it can only be passed by a student who has reached the level, and can be passed by any student who has. Most unfair checks fail the first half: they can be passed by recall when the objective asked for judgement.

WHAT EACH LEVEL CAN BE CHECKED WITH

- **Level 1** — multiple choice, and short answers that ask for a list, a name or a definition. Many short questions, spread out. Long-answer marks spent here are wasted, because a long answer cannot distinguish a student who knows the list from one who knows it better.
- **Level 2** — short answers whose stem requires the material rearranged: sorted, compared, put in the student's own sequence. **Test your stem this way:** if it can be answered word-for-word from the notes, it has slipped to level 1 whatever you intended.
- **Level 3** — a short or structured long answer built on a case the student has not seen. If the case is familiar, the question measures memory.
- **Level 4** — structured questions that ask for the parts and for how the parts relate.
- **Level 6** — long answers. Give the marks for the reasoning and the criteria, not for the facts repeated inside the answer. A student who reaches a sound conclusion for stated reasons has met the level; one who reaches it by recalling what the book concluded has not.

DOING OBJECTIVES CAN ONLY BE CHECKED BY WATCHING

This is worth stating plainly, because the curriculum's prescribed tool for practical objectives is very often a viva. A viva is a good tool. It shows whether a student knows a procedure and can explain it. It cannot show whether they can carry it out, and a fluent student who has never handled the material will pass it comfortably.

Where an objective is about doing, the evidence has to be seen: a checklist you tick while the student works, an OSPE station, or an observed performance. This need not be elaborate. A checklist of the steps you will look for, written before the session, is enough — and it is the only record that shows the procedure was actually performed.

ATTITUDE OBJECTIVES NEED EVIDENCE OVER TIME

Conduct across the term, a short written reflection, a checklist applied on more than one occasion. Decide in advance when you will collect it. An affective objective that reaches the end of term uncollected has not been assessed, however carefully it was written.

> **Three questions before you set any paper.** First: for each question, what level does it actually reach — not what level was intended? Second: could a student pass it by recall alone? Third: does the balance of marks match the balance of the syllabus, or have the easy-to-mark levels quietly taken the marks? Appendix C6 gives the level and priority profile of every Unit, which is the fastest way to answer the third.

PART A · CHAPTER 8

How to use a Unit in Part B

Part B has thirty-six Units: twenty-six theory Units following the topics of Papers 1 and 2, and ten practical Units following Table 4. Each begins on its own page, and each is laid out the same way, so that once you have read one you can use any of them.

WHAT IS ON THE PAGE

1. **The heading strip** — hours, term, marks, how many objectives, and the highest level the Unit asks for. Read the last of these first. It tells you whether this is a Unit that stops at recall or one that reaches judgement, and that single fact shapes everything else.
2. **Four counters** — how many objectives fall in each domain, how many are Must know, and which kind of knowledge dominates.
3. **What this topic asks students to do** — the levels as a picture, in both schemes side by side. Empty levels are shown as empty rather than left out, because knowing what a topic does *not* ask for is as useful as knowing what it does.
4. **What kind of knowledge is involved** — the four kinds, with how each is learned.
5. **The objectives, grouped by level** — the curriculum's own objectives, sorted into their levels so that a session can be planned around one level at a time. The last column tells you how to pitch each one.
6. **Teaching and checking, level by level** — what the teaching has to accomplish at each level this Unit uses, and what a fair check looks like.
7. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Go to the grouped objectives and take one level at a time. Teach upward: the lower levels are the ground the higher ones stand on. Then check the method against Chapter 6, and the check you have in mind against Chapter 7.

Setting a paper. Use the level picture to decide where the marks should go, and the grouped objectives to see which ones can carry a long question. A Unit whose highest level is Comprehension cannot honestly carry a question that asks for judgement, however many marks it is worth in the syllabus.

Giving students their bearings. The student panel is written to be read by students, not about them. The single most useful thing in it is the first line: read the verb in every objective. Students who do that stop preparing for the wrong question.

> **What this book does not do.** It does not rewrite the curriculum, and it does not tell you what to teach in Rasashastra and Bhaishajya Kalpana — that is your subject, and every objective in Part B appears in the curriculum's own words. What the book adds is the taxonomy layer: what kind of learning each objective asks for, and what follows for teaching and checking. Where the wording of the curriculum and its own code point in different directions, the Unit shows you both and leaves the decision with you.

PART B

The Units

Thirty-six Units, each on its own pages: twenty-six following the theory topics of Papers 1 and 2, and ten following the practical topics of Table 4. Every objective appears in the curriculum's own words, sorted into its level so that a session can be planned around one level at a time. Chapter 8 explains the layout; after that the Units are meant to be opened at whichever one you are about to teach.

1. **Paper 1** — Ayurvediya Aushadhi Nirmana Vigyana, Units 1 to 14
2. **Paper 2** — Ayurvediya Aushadhi Prayoga Vigyana, Units 1 to 12
3. **Practical** — NCISM Table 4, Units 1 to 10

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PAPER 1 · UNIT 1

Chronological development of Ayurvediya Aushadhi Nirmana

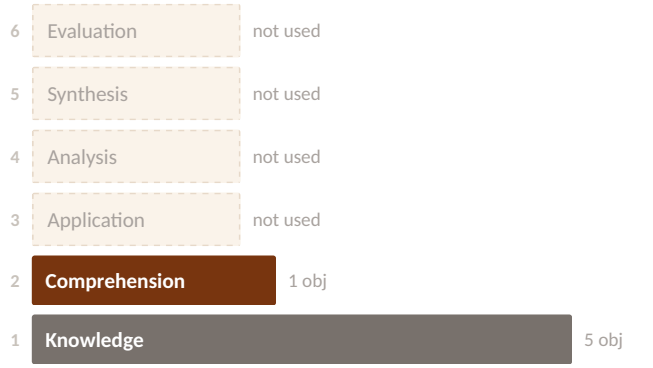
Hours: 2 lecture · 1 non-lecture | Term 1 | Theory marks: 5 | **7 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**1**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	3 desirable · 1 nice to know	kind of knowledge

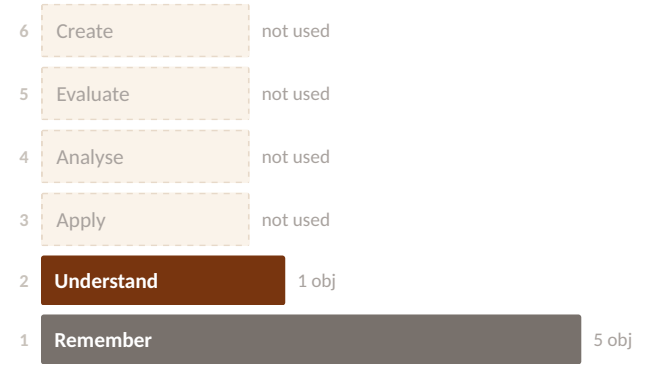
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**5**	**1**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 5 OBJECTIVES				
Explain historical evolution of Ayurvediya aushadhi nirman and Rasashastra.	*explain*	Conceptual ◦	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Describe about contribution of Nagarjuna Acharya to Rasashastra.	*describe*	Conceptual ◦	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist important classical texts of Rasashastra and describe their unique features in short.	*enlist*	Conceptual ◦	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Describe Recent development in Ayurvedic Pharmaceuticals viz new dosage forms, pharmaceutical modification techniques.	*describe*	Procedural °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Define Rasa and Rasayana and describe difference between Rasa and Rasayana.	*define*	Factual °	Desirable	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Describe structure of Pharmacy and enlist formulations prepared in pharmacy, after visiting the unit of teaching pharmacy of own campus.	*describe*	Conceptual °	Must know	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Justify design of ancient Rasashala.	*justify*	Conceptual	Nice to know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	5	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 6 of these 7 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 2

Paribhasha (Terminology)

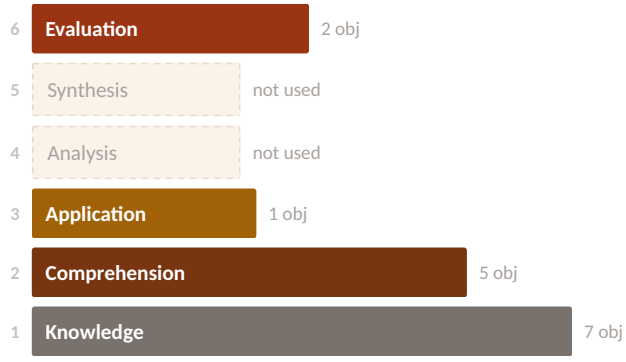
Hours: 8 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **19 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
15	**4**	**12**	**Factual**
levels 1–6	Krathwohl domain	5 desirable · 2 nice to know	kind of knowledge

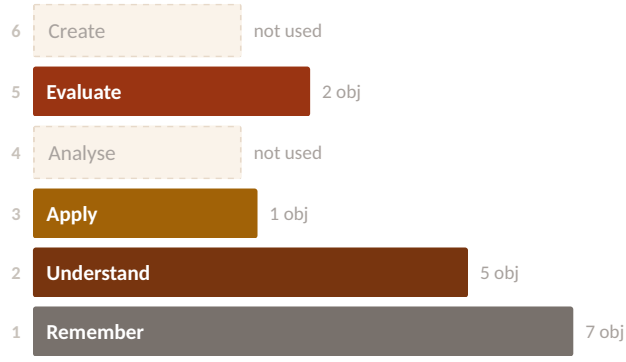
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
11	**8**	**3**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 7 OBJECTIVES				
Explain the term Paribhasha and its importance in Ayurvediya Aushadhi Nirmana.	*explain*	Factual + Conceptual °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Enlist and discuss Sandigdha, Krutrima, Pratinidhi and Anupalabdha dravya.	*enlist*	Factual °	Desirable	Teach at this level.
Explain the term Marana and describe its types with examples.	*explain*	Factual + Conceptual °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Enlist all relevant prakriya paribhasha of Amrutikarana Lohitikarana, Sattvapatana, druti and discuss with examples.	*enlist*	Factual °	Desirable	Teach at this level.
Identify names of Parada Bandhas.	*identify*	Factual	Nice to know	Teach at this level.
Describe Grahyagrahya parameters used for selection of Rasadravyas.	*describe*	Procedural °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain importance of Mana-paribhasha and classical types of Mana. Recall Charts of Mana.	*explain*	Conceptual	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 5 OBJECTIVES				
Describe the terms Aushadha, Bheshaja, Kalpana, Kashaya Kashaya yoni, Samskara- based on their word derivations.	*describe*	Factual °	Must know	Teach at this level.
Enlist sequentially - names of all drugs classified in the varga(group). Recite shlokas of Maharasa, Uprasa, Sadharana Rasa Varga from Rasaratnasamuchchaya.	*enlist*	Factual	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Recite sequentially names of Parada Ashta samskara. Explain the terms Jarana Murchchhana and cite types with examples. Compare Jarana and Murchchhana.	*recite*	Factual	Must know	Teach at **level 1**. The verb *recite* is what the exam will ask for.
Illustrate all Bhasma pariksha as per classical description.	*illustrate*	Conceptual °	Must know	Teach at this level.
Recognise and discuss Dhanvantari Bhaga and Rudra bhaga.	*recognise*	Factual °	Nice to know	Teach at **level 1**. The verb *recognise* is what the exam will ask for.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Describe the definition of the term Shodhana. Explain with examples different techniques used for the procedures of Shodhana.	*describe*	Factual + Procedural °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 6 · EVALUATION / EVALUATE — 2 OBJECTIVES				
Define the word Siddhilakshani. Recite examples of classical siddhilakshani. Interpret its rationality.	*define*	Factual °	Must know	Teach at **level 1**. The verb *define* is what the exam will ask for.
Categorize parameters of Drug Standardization and develop a checklist for assessment of quality of rasadravyas.	*categorize*	Procedural	Desirable	Teach at **level 4**. The verb *categorize* is what the exam will ask for.
AFFECTIVE DOMAIN — 4 OBJECTIVES				
Explain Value of selection of genuine raw material.	*explain*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Explain importance of ethical practices for drug processing(Shodhan, Marana).	*explain*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Discuss and justify importance of Bhasma Pariksha.	*discuss*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Explain value of keen and accurate application of weights and measures in Ayurvediya Aushadhi nirmana.	*explain*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	7	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	5	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	2	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 12 of these 19 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Factual* — recall it until it sticks; *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PAPER 1 · UNIT 3

Adharbhuta Siddhanta (Application of fundamental principles)

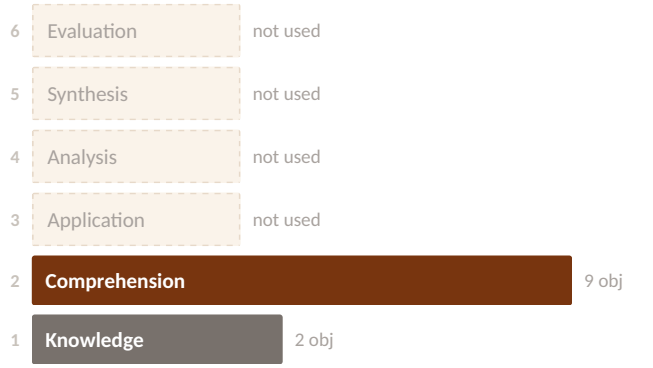
Hours: 4 lecture · 2 non-lecture | Term 1 | Theory marks: 5 | **11 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
11	**0**	**9**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 0 nice to know	kind of knowledge

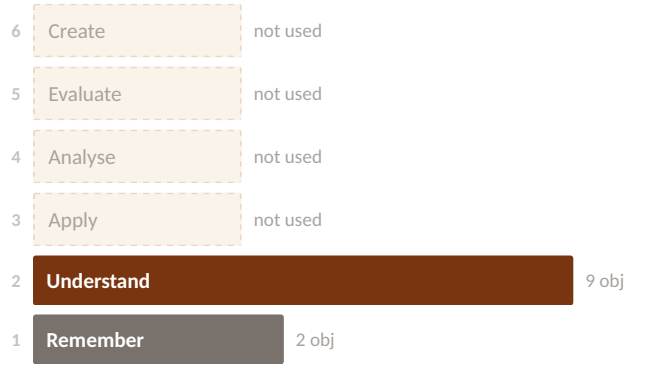
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**10**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Explain Saveeryata Avadhi(Shelf life) of Ayurvedic dosage forms.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Explain importance of Kala (Time) Samskara in Ayurvediya Aushadhi Nirmana. Elaborate Aushadhi sevana kala mentioned in Sharangdhara samhita. Describe chrono- Therapeutics.	*explain*	Conceptual	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 9 OBJECTIVES				
Elaborate fundamental principles of Ayurvediya Aushadhi Nirmana alongwith their classical references and discuss their application with classical examples of various kalpas.	*elaborate*	Conceptual	Must know	Teach at this level.
Explain Dravya Samgraha vidhi. Explain types of Bhumi desha and types of drugs to be collected from particular place & places from where Dravya should not be collected.	*explain*	Conceptual	Must know	Teach at this level.
Describe time of Dravya collection and explain rationality behind.	*describe*	Conceptual °	Must know	Teach at this level.
Enlist parts of Sthavara & Jangama Dravyas used for Ayurvediya Aushadhi nirmana.	*enlist*	Factual °	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Define Samskara. Discuss role of Samskara in Ayurvediya Aushadhi Nirmana alongwith various examples mhita.	*define*	Conceptual °	Must know	Teach at **level 1**. The verb *define* is what the exam will ask for.
Describe Ardra - Shushka Dravya & Anukta – Visheshokta Dravya collection principles.	*describe*	Conceptual	Must know	Teach at this level.
Describe importance of kalpa sevan matra. Explain factors considered for deciding dosage of any drug (Ayurvedic as well as modern medicine principles). Describe Posology.	*describe*	Conceptual	Must know	Teach at this level.
Discuss Yogika Dravya Sidhdhanta(Drug combination).	*discuss*	Conceptual °	Must know	Teach at this level.
Justify Aushadhi kalpa namakarana siddhanta with examples.	*justify*	Conceptual °	Desirable	Teach at **level 6**. The verb *justify* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	9	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 11 of these 11 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 4

Yantropakaranani- I (Equipments and machineries)

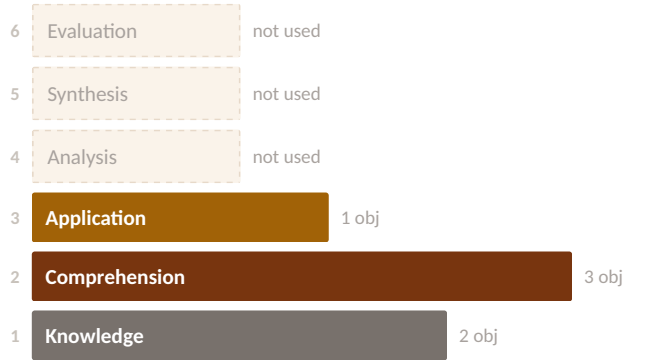
Hours: 6 lecture · 4 non-lecture | Term 1 | Theory marks: 5 | **6 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**0**	**1**	**Conceptual**
levels 1–6	Krathwohl domain	5 desirable · 0 nice to know	kind of knowledge

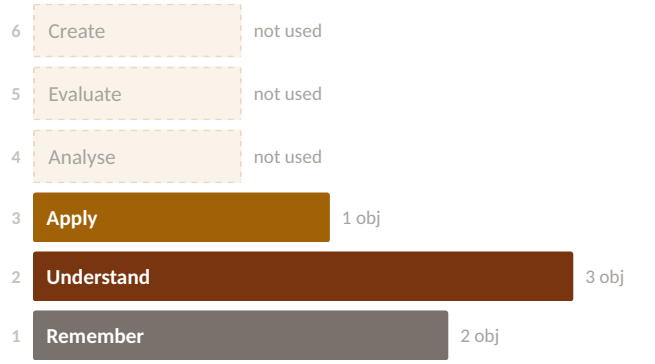
WHAT THIS TOPIC ASKS STUDENTS TO DO

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BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



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WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**3**	**2**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Enlist categorical information about the following Yantras in the charts- Ulukhal Yantra, Patan Yantra, Jarana Yantra, Patala Yantra and Swedana Yantra.	*enlist*	Factual	Desirable	Teach at this level.
Enlist Ayurvediya aushadhi kalpana and equipments/yantras/ machines used for preparation of each kalpana.	*enlist*	Procedural °	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 3 OBJECTIVES				
Choose and record contemporary machines used in Ayurvedic drug preparation.	*choose*	Procedural °	Desirable	Teach at **level 4**. The verb *choose* is what the exam will ask for.
Discuss the pharmaceutical use of Distillation apparatus, Ball Mill, Pulveriser, End Runner, Edge Runner, Tablet compression machine, Capsule filling machine, Pouch filling machine, Liquid filling machine in Ayurvediya Aushadhi Nirmana.	*discuss*	Conceptual °	Desirable	Teach at this level.
Interpret the mechanism and effect of Yantras / machines on the Physical and Chemical properties of the drug material.	*interpret*	Conceptual °	Desirable	Teach at **level 6**. The verb *interpret* is what the exam will ask for.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Describe the principles behind construction and working of the classical Yantras used for Ayurvediya Aushadhi nirmana.	*describe*	Conceptual	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	3	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 5 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 1 · UNIT 5

Yantropakaranani -II (Equipments, fuel and Heating Devices)

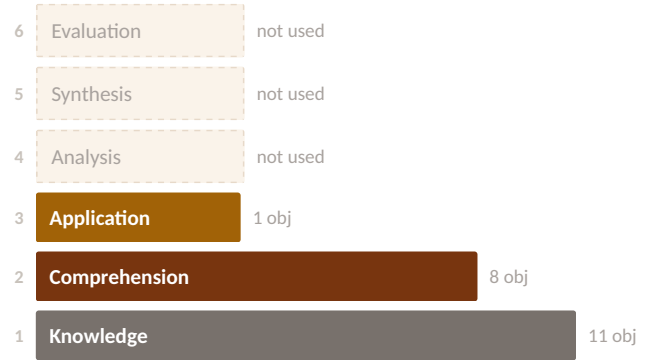
Hours: 5 lecture · 4 non-lecture | Term 1 | Theory marks: 5 | **21 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
20	**1**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	14 desirable · 4 nice to know	kind of knowledge

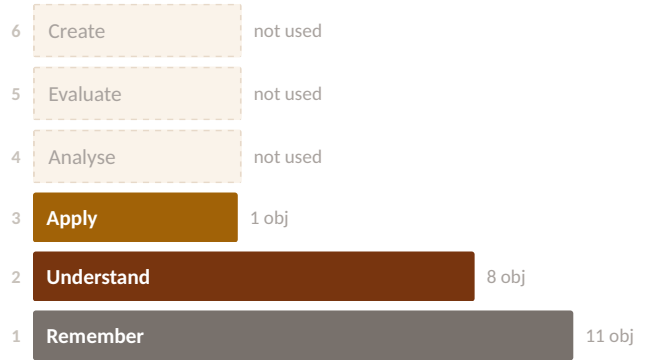
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
9	**11**	**3**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 11 OBJECTIVES				
Describe the term Yantra and enlist yantras described in classics useful for aushadhi nirmana.	*describe*	Factual ◦	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Define the term Puta and recite its classical explanation.	*define*	Factual	Must know	Teach at this level.
Produce categorical information(Size of Pit, Number of cowdunkcakes, use etc.) about following Putas viz. Mahaputa Gajaputa, Varahputa, Kukkutaputa, Kapotputa, Lavakputa Kumbhaputa, Bhandaputa, Valukaputa and Bhudharputa.	*produce*	Procedural ◦	Must know	Teach at **level 5**. The verb *produce* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Compile names and significance of temperature measuring devices with reference to Puta.	*compile*	Factual + Conceptual °	Desirable	Teach at **level 4**. The verb *compile* is what the exam will ask for.
Enlist various devices used for heating during Ayurvediya Aushadhi nirmana.	*enlist*	Factual °	Desirable	Teach at this level.
Enlist constituents needed for Samanya Musha Nirmana.	*enlist*	Factual °	Desirable	Teach at this level.
Define the term Musha and describe uses of various types of Musha.	*define*	Factual + Conceptual °	Desirable	Teach at this level.
Enlist and discuss the material used for Sandhi Bandhan (while using Sharava and Kachakupi).	*enlist*	Factual °	Nice to know	Teach at this level.
Identify and record the types of Crucibles.	*identify*	Conceptual °	Nice to know	Teach at this level.
Describe types of Koshthi and their uses.	*describe*	Conceptual	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist various fuels used for heating for estimation of their heat value.	*enlist*	Conceptual °	Nice to know	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 8 OBJECTIVES				
Recognise the principles behind construction and working of the classical Putas.	*recognise*	Conceptual °	Must know	Teach at **level 1**. The verb *recognise* is what the exam will ask for.
Identify and enlist contemporary devices used in the preparation of Bhasma. T.	*identify*	Procedural °	Desirable	Teach at **level 1**. The verb *identify* is what the exam will ask for.
Review the temperature pattern of various Putas by referring research articles.	*review*	Conceptual °	Desirable	Teach at **level 4**. The verb *review* is what the exam will ask for.
Enlist the specific Puta used for a particular Bhasma Nirmana.	*enlist*	Factual °	Desirable	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Recall Pakaj Utpatti Siddhant and interpret it for Agni Sannikarsha Sanskara.	*recall*	Factual °	Desirable	Teach at **level 1**. The verb *recall* is what the exam will ask for.
Explain the Pharmaceutical use of the Hot plate, heating mantle induction stove, hot air oven, muffle furnace (horizontal and vertical type).	*explain*	Conceptual °	Nice to know	Teach at this level.
Describe the Current trends in heat transfer device e.g. steam jacketed heating device, programmed muffle furnace programmed furnace for Parpati / Pottali preparation.	*describe*	Procedural °	Desirable	Teach at this level.
Assess and interpret the effect of fuel in quantum of heat given (time and temperature).	*assess*	Conceptual °	Desirable	Teach at **level 6**. The verb *assess* is what the exam will ask for.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Interpret the effect of heat transformation in the material subjected to heating through Musha.	*interpret*	Conceptual °	Desirable	Teach at **level 6**. The verb *interpret* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Analyse and appraise use of proper heating device and fuels used for Ayurvediya Aushadhi Nirmana.	*analyse*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	11	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	8	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 19 of these 21 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 1 · UNIT 6

Kalpana Nirmana I (Primary & Secondary dosage forms)

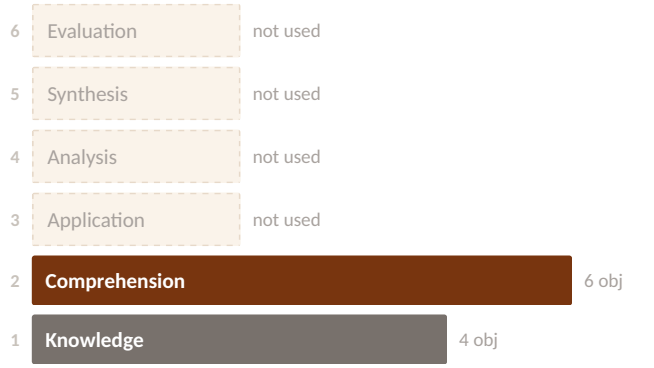
Hours: 6 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **11 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
10	**1**	**7**	**Conceptual**
levels 1–6	Krathwohl domain	4 desirable · 0 nice to know	kind of knowledge

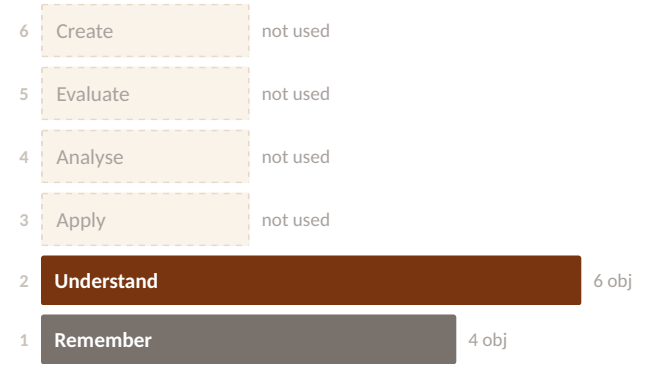
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
3	**7**	**3**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 4 OBJECTIVES				
Define with synonyms and classify different varieties of the dosage form in Ayurvediya Aushadhi nirmana.	*define*	Factual ◦	Must know	Teach at this level.
Describe method of preparation of dosage form along with the principle involved.	*describe*	Conceptual + Procedural ◦	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Describe meaning of the term Upakalpana. Enlist panchavidha kashaya kalpana and their respective Upakalpana.	*describe*	Factual ◦	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Explain various kalpas belonging to Various dosage forms - Ingredients, proportion, matra, sevan vidhi if any, therapeutic.	*explain*	Conceptual °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 6 OBJECTIVES				
1. Explain Kashaya kalpana 2. Enlist the dosage form come under Panchavidha Kashaya Kalpana and their Upakalpana 3. Justify Primary, secondary and tertiary dosage form under Panchavidha kashaya Kalpana and their Upakalpana.	*explain*	Conceptual °	Must know	Teach at this level.
Describe methods of preparation of the dosage form along with the principles of extraction, concentration and dilution etc.	*describe*	Conceptual + Procedural °	Must know	Teach at this level.
Explain in details about Standard Operating Procedure (SOP) of each kalpana.	*explain*	Procedural	Must know	Teach at this level.
Enlist details of the applications/administration (samanya Matra Anupana or sahapana, indications and contraindications) of the dosage form with various examples.	*enlist*	Factual °	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Assess and discuss modern day development and market trend of the dosage form.	*assess*	Conceptual °	Desirable	Teach at **level 6**. The verb *assess* is what the exam will ask for.
Determine the advantages and disadvantages of the dosage form.	*determine*	Conceptual °	Desirable	Teach at **level 4**. The verb *determine* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Critically evaluate rationality behind different proportion of water used in various formulations.	*critically*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	4	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	6	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 10 of these 11 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 7

Kalpana Nirmana-II (Method of Preparation of different dosage forms& Dietary Supplements))

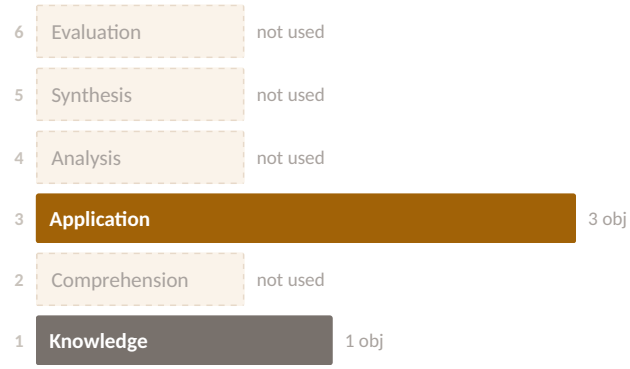
Hours: 5 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **4 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
4	**0**	**4**	**Procedural**
levels 1–6	Krathwohl domain	0 desirable · 0 nice to know	kind of knowledge

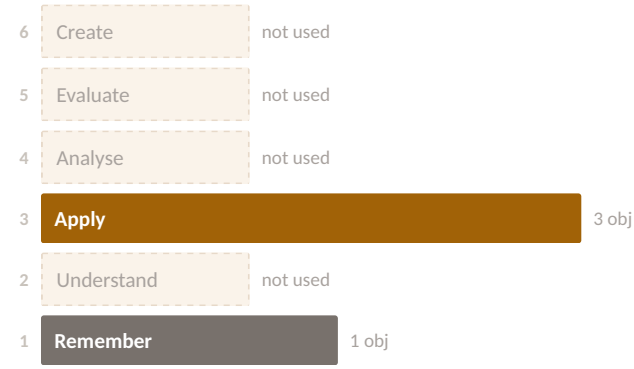
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**3**	**4**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Sneha Kalpana, Kalpana with examples instruments used in small and Large scale production, Concept of Avartana, Research updates on Snehakalpana, Market survey.	*explain*	Conceptual + Procedural °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Avaleha, Kalpana with examples, instruments used in small and Large scale production, Research updates.	*explain*	Procedural	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Describe the Significance of sandhana kalpana, classification Difference between Madya and Shukta Kalpana, general method of preparation, essential ingredients, Anukta Mana, Sandhana Vidhi, observations, Burning candle test, Lime water Test difference between Asava and Arishta, essential knowledge of sale and clinical practice, Research updates.	*describe*	Conceptual + Procedural °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain Definition, significance of Pathya, types, general method of preparation Manda, Peya, Yavagu, Vilepi, Anna or Odana Kalpana, Krushara, Yusha, Takra, Khada, Kambalika Raga, Shadava, Related Research updates, Market survey of Dietary Supplements and Nutraceuticals.	*explain*	Conceptual + Procedural °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	3	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 1 of these 4 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Procedural* — do it, do not only read it; *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 1 · UNIT 8

Rasa Dravya Parichaya- I

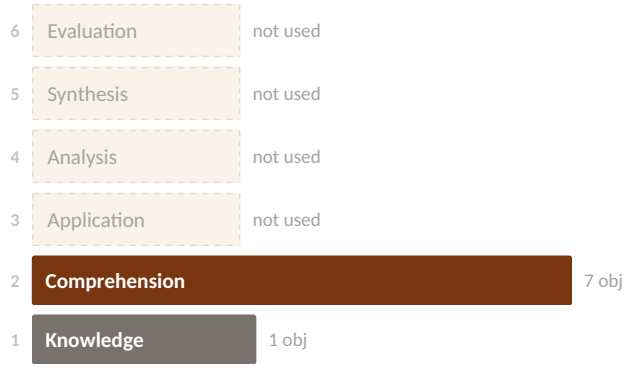
Hours: 12 lecture · 4 non-lecture | Term 2 | Theory marks: 10 | **9 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
8	**1**	**7**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 0 nice to know	kind of knowledge

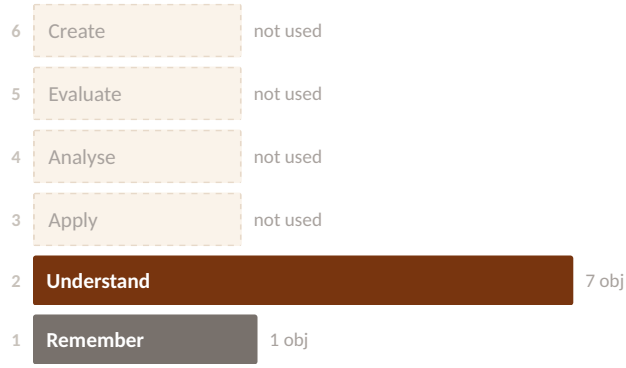
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**5**	**4**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Enlist Kalpas prepared from these Rasdravyas and therapeutic importance of that Rasadravya.	*enlist*	Conceptual ◦	Desirable	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 7 OBJECTIVES				
Enlist synonyms of Rasadravya and explain significance of that T.	*enlist*	Conceptual ◦	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Explain classical & Mineralogical identification and Sources of Rasadravyas.	*explain*	Conceptual ◦	Must know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Discriminate types, Grahya-Agrahyata and Dosha of Rasadravya.	*discriminate*	Conceptual	Must know	Teach at **level 4**. The verb *discriminate* is what the exam will ask for.
Explain Shodhan and Marana and other processing techniques of the Rasa- Dravya.	*explain*	Procedural °	Must know	Teach at this level.
Discuss probable physical, chemical changes occurring during process & after Shodhana and Marana of Parada, Haratala Tamra and Shankha.	*discuss*	Procedural °	Must know	Teach at this level.
Explain research updates about Shilajatu, Naga, Mukta and Kapardika.?	*explain*	Conceptual °	Desirable	Teach at this level.
Discuss probable physical, chemical changes occurring after Bhavana, Puta & whole process of Bhasma nirmana from Rasadravyas.	*discuss*	Procedural °	Must know	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Develop checklist for identification of genuine rasadravyas.	*develop*	Procedural	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	7	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 8 of these 9 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 9

Rasa Dravya Parichaya II

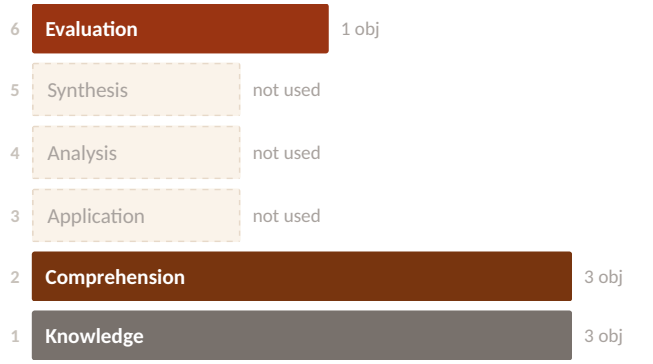
Hours: 7 lecture · 6 non-lecture | Term 2 | Theory marks: 5 | **7 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
7	**0**	**1**	**Conceptual**
levels 1–6	Krathwohl domain	6 desirable · 0 nice to know	kind of knowledge

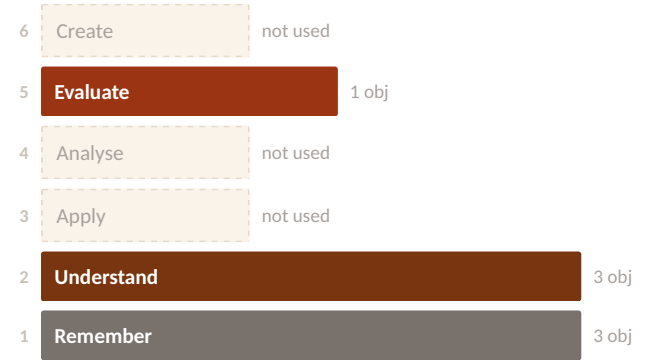
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**4**	**2**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Explain classical & Mineralogical identification and their Sources.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Elaborate types, Grahya-Agrahyata and Dosha of Rasadravyas.	*elaborate*	Conceptual	Desirable	Teach at **level 2**. The verb *elaborate* is what the exam will ask for.
Enlist names of rasadravyas and important kalpas prepared from respective dravya. Describe therapeutic importance of that Rasadravya.	*enlist*	Factual + Conceptual °	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 3 OBJECTIVES				
Enlist the important synonyms of Rasadravyas.	*enlist*	Factual °	Desirable	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Explain Shodhana, Marana & and other processing techniques with probable chemical reactions.	*explain*	Procedural °	Must know	Teach at this level.
Explain research updates of Kasisa, Gauripashana and Akika.	*explain*	Conceptual °	Desirable	Teach at this level.
LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Explain Shodhana, Marana & and other processing techniques with probable chemical reactions.	*explain*	Procedural °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	3	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	1	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 6 of these 7 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Evaluation* — decide something, write down the reasons you used, then argue the opposite side.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PAPER 1 · UNIT 10

Rasadravya Parichaya III

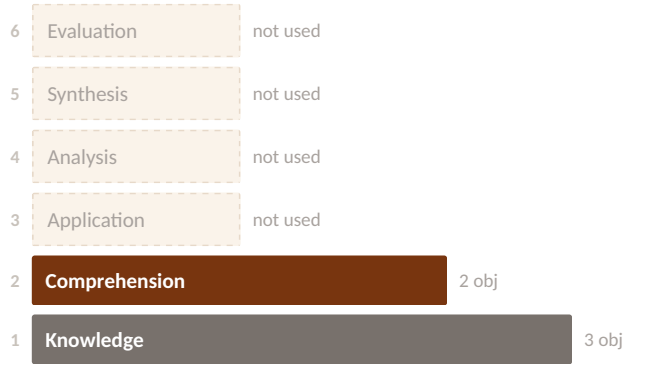
Hours: 3 lecture · 6 non-lecture | Term 2 | Theory marks: 5 | **5 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
5	**0**	**0**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 3 nice to know	kind of knowledge

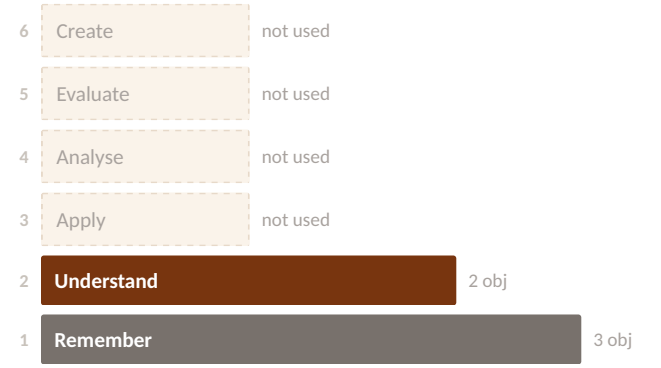
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**3**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Enlist Synonyms and sources.	*enlist*	Factual ◦	Nice to know	Teach at this level.
Explain research update of Badarashma.	*explain*	Conceptual ◦	Nice to know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Enlist Rasadravyas from this group and describe important kalpas with their therapeutic utility.	*enlist*	Factual ◦	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Determine types, Grahya-Agrahya, properties & classical as well as Mineralogical Identification of Rasadravyas.	*determine*	Conceptual	Nice to know	Teach at **level 4**. The verb *determine* is what the exam will ask for.
Explain Shodhana, Marana and Probable Chemical Changes.	*explain*	Conceptual °	Desirable	Teach at this level.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 5 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 11

Kalpana Nirman -III (Method of Preparation of different dosage forms)

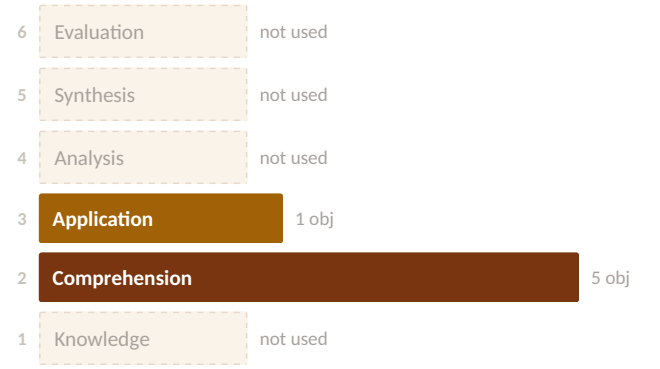
Hours: 8 lecture · 6 non-lecture | Term 2 | Theory marks: 10 | **6 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**0**	**2**	**Procedural**
levels 1–6	Krathwohl domain	1 desirable · 3 nice to know	kind of knowledge

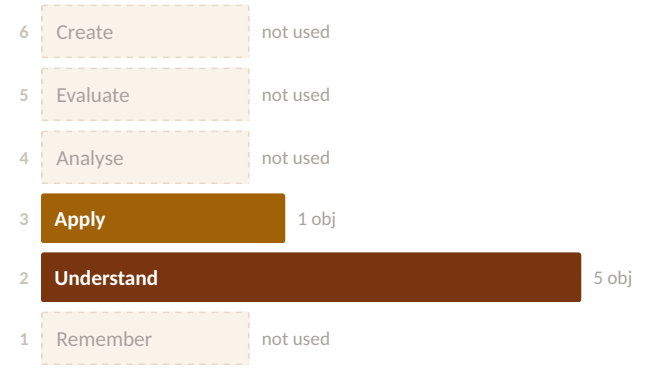
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**2**	**6**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 5 OBJECTIVES				
Describe sharkara kalpana along with preparation method of Syrup. Explain therapeutic importance of prepared sharkara kalpana along with its shelf life.	*describe*	Conceptual + Procedural °	Nice to know	Teach at this level.
Describe Gudapaka preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Gudapaka along with its shelf life.	*describe*	Conceptual + Procedural °	Nice to know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Describe Lavana kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lavana kalpana along with its shelf life and packaging techniques.	*describe*	Procedural	Desirable	Teach at this level.
Describe Ayaskriti kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Ayaskriti kalpana along with its shelf life and packaging techniques.	*describe*	Procedural	Nice to know	Teach at this level.
Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.	*describe*	Procedural	Must know	Teach at this level.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Describe Kshara kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Kshara kalpana and Ksharasutra integration with Shalya tantra department along with its shelf life and packaging techniques.	*describe*	Procedural	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	—	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	5	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 5 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Procedural* — do it, do not only read it; *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 1 · UNIT 12

Chaturvidha Rasayana

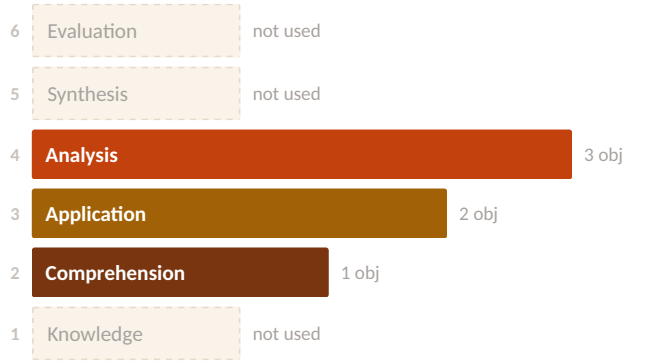
Hours: 4 lecture · 4 non-lecture | Term 2 | Theory marks: 10 | **6 objectives** | Highest level asked for: **Analysis / Analyse**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	0	2	Conceptual
levels 1–6	Krathwohl domain	4 desirable · 0 nice to know	kind of knowledge

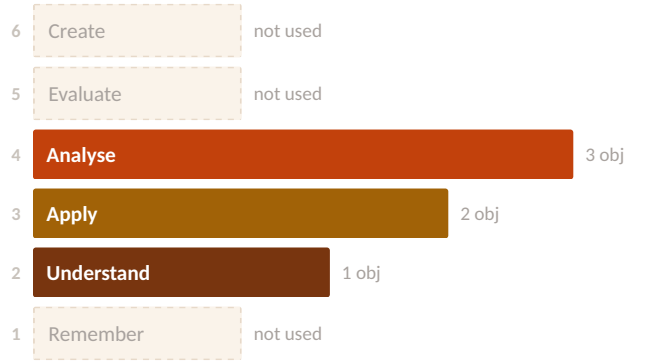
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	4	4	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Describe importance and unique features of Chaturvidha Rasayana - Kharaliya Rasayana, Parpati Rasayana, Kupipakwa Rasayana & Pottali Rasayana.	*describe*	Conceptual	Must know	Teach at this level.
LEVEL 3 · APPLICATION / APPLY — 2 OBJECTIVES				
Explain definition, types and method of preparation of Chaturvidha Rasayana.	*explain*	Conceptual + Procedural °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Explain law of Definite proportion and proportion of Mercury and sulphur needed for preparation of sulphide of Mercury. in Chaturvidha rasayana.	*explain*	Procedural °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
LEVEL 4 · ANALYSIS / ANALYSE — 3 OBJECTIVES				
Determine role of Yantra, Agni & Kala for pharmaceutical process of chaturvidha rasayana.	*determine*	Conceptual + Procedural °	Desirable	Teach at this level.
Explain shelf-life of Chaturvidha Rasayana.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Describe chemical changes occurring during the preparation of chaturvidha rasayana and its impact on Pharmacological action of Chaturvidha Rasayana.	*describe*	Procedural °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	—	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	2	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	3	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 1 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it; *Analysis* — put two things side by side and write down what differs, and why.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 4, repeating it does not mean you know it.

PAPER 1 · UNIT 13

Current and emerging trend in Ayurvedic pharmaceuticals

Hours: 3 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Highest level asked for: **Knowledge / Remember**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
5	**1**	**0**	**Conceptual**
levels 1–6	Krathwohl domain	5 desirable · 1 nice to know	kind of knowledge

WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3

6	Evaluation	not used
5	Synthesis	not used
4	Analysis	not used
3	Application	not used
2	Comprehension	not used
1	**Knowledge**	5 obj

REVISED 2001 — Anderson & Krathwohl

6	Create	not used
5	Evaluate	not used
4	Analyse	not used
3	Apply	not used
2	Understand	not used
1	**Remember**	5 obj

The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**4**	**1**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 5 OBJECTIVES				
Discuss the classification of different dosage forms.	*discuss*	Conceptual	Desirable	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Explain the need of different dosage forms.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Enlist categorical information about the cosmetics used in daily routine. Appreciate Ayurvedic cosmetic products.	*enlist*	Factual °	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Choose and record the contemporary machineries used in the manufacture of cosmetics.	*choose*	Conceptual °	Nice to know	Teach at **level 4**. The verb *choose* is what the exam will ask for.
Enlist Quality Control parameters of cosmetics preparations.	*enlist*	Procedural °	Desirable	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Appreciate Ayurvedic cosmetic products.	*appreciate*	Conceptual °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	5	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 5 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
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FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 1, repeating it does not mean you know it.

PAPER 1 · UNIT 14

GMP(Schedule T) & Regulatory aspects of Ayurvedic drugs

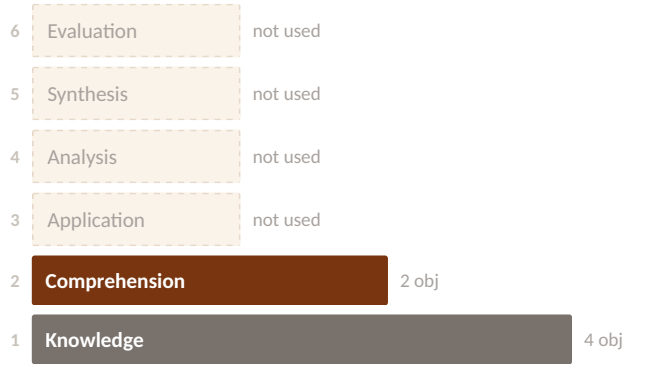
Hours: 2 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **7 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**1**	**2**	**Conceptual**
levels 1–6	Krathwohl domain	4 desirable · 1 nice to know	kind of knowledge

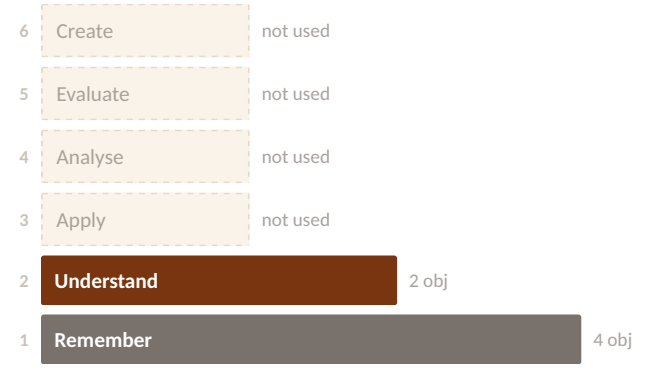
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**6**	**1**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 4 OBJECTIVES				
Discuss the guidelines of Food Safety and Standards Authority of India (FSSAI) and FDA.	*discuss*	Conceptual °	Desirable	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Determine the principles and practice of establishment of Ayurvedic pharmacy.	*determine*	Conceptual	Desirable	Teach at **level 4**. The verb *determine* is what the exam will ask for.
Discuss the NABL guidelines for testing laboratory (Chemical).	*discuss*	Procedural °	Nice to know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Explain long forms of these - (FSSAI) and (FDA), (CCRAS), (DCGI), (CDSCO).	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Explain the legal and regulatory aspects of manufacturing, and sale of Ayurvedic drugs.	*explain*	Conceptual °	Must know	Teach at this level.
Describe acts and rules mentioned in Drug & Cosmetic Act 1940 & Rule 1945 and their relevance to Ayurvedic, Siddha, Unani (ASU) drugs.	*describe*	Conceptual °	Must know	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Discuss ethical aspect of large scale drug preparation in Ayurvedic Pharmacies.	*discuss*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	4	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 6 of these 7 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 2 · UNIT 1

Aushadhi Prayoga Vigyana

Hours: 1 lecture · 2 non-lecture | Term 1 | Theory marks: 5 | **1 objectives** | Highest level asked for: **Knowledge / Remember**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
1	**0**	**1**	**Factual**
levels 1–6	Krathwohl domain	0 desirable · 0 nice to know	kind of knowledge

WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3

6	Evaluation	not used
5	Synthesis	not used
4	Analysis	not used
3	Application	not used
2	Comprehension	not used
1	**Knowledge**	1 obj

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6	Create	not used
5	Evaluate	not used
4	Analyse	not used
3	Apply	not used
2	Understand	not used
1	**Remember**	1 obj

The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	—	—	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Define Aushadhi Prayoga Vigyana and its scope and enlist prashasta bheshaja laxana.	*define*	Factual °	Must know	Teach at this level.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 1 of these 1 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 1, repeating it does not mean you know it.

PAPER 2 · UNIT 2

Single drug (Herbal & Mineral)

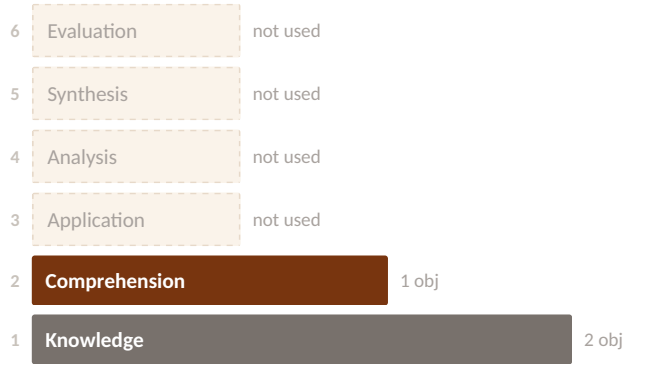
Hours: 8 lecture · 2 non-lecture | Term 1 | Theory marks: 10 | **5 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
3	**2**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	1 desirable · 1 nice to know	kind of knowledge

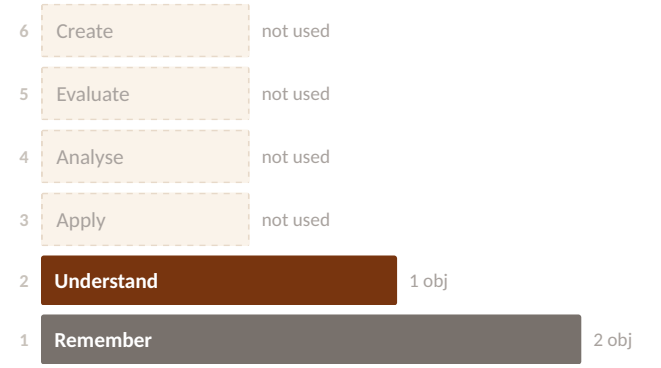
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



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The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**5**	**1**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Describe different dosage forms prepared out of a single herb or mineral.	*describe*	Conceptual ◦	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Describe therapeutic efficacy of different formulations of Vishadrvya (Bhallataka), with its toxic effects and remedy.	*describe*	Conceptual ◦	Nice to know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Describe therapeutic efficacy of different formulations of Guduchi, Gairika, Gandhaka.	*describe*	Conceptual ◦	Must know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
AFFECTIVE DOMAIN — 2 OBJECTIVES				
Appreciate how the therapeutic efficacy varies depending on the dosage form.	*appreciate*	Conceptual °	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Appraise multiple factors considered for preparation of various dosage forms from a single drug and creat list of more such examples.	*appraise*	Factual + Conceptual + Procedural °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
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- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 2 · UNIT 3

Single drug(Bhasma, Shuddha & Pishti)

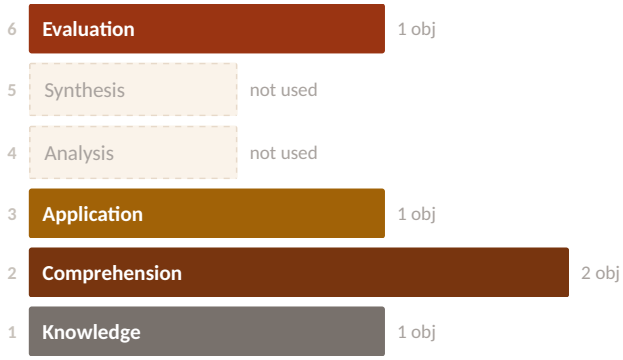
Hours: 12 lecture · 6 non-lecture | Term 2 | Theory marks: 15 | **5 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
5	**0**	**2**	**Conceptual**
levels 1–6	Krathwohl domain	3 desirable · 0 nice to know	kind of knowledge

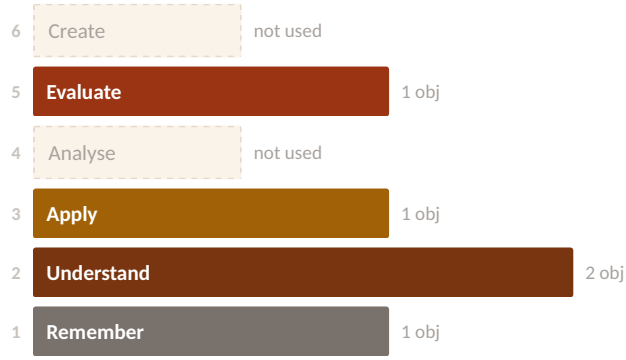
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

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FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**5**	—	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Describe Apakwa Ashuddha Avidhi Bhasma Sevanajanya Vyadhi and their Shanti-upaya.	*describe*	Conceptual ◦	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Explain chemical form/composition, Pharmacodynamics and pharmacokinetics of Abhraka, Loha and Godanti Bhasma.	*explain*	Conceptual ◦	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Explain therapeutic properties, dosage, Anupana, Pathyapathya duration of treatment, Sevana Kala, shelf life, important Yogas of each Bhasma.	*explain*	Conceptual °	Must know	Teach at this level.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Describe in detail Amayika Prayoga (Therapeutic uses) of each Bhasma/Pishti with given references.	*describe*	Conceptual °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Explain research updates and clinical evidences of Swarna Bhasma, Makshika Bhasma and Shankha Bhasma.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	1	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PAPER 2 · UNIT 4

Aushadhi Kalpa -I (Compound formulations)

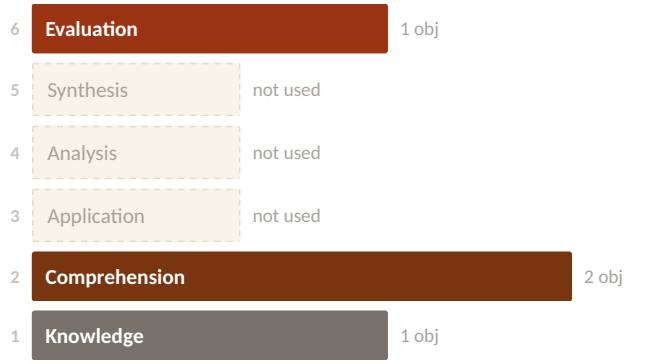
Hours: 16 lecture · 4 non-lecture | Term 2 | Theory marks: 15 | **5 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
4	**1**	**2**	**Conceptual**
levels 1–6	Krathwohl domain	3 desirable · 0 nice to know	kind of knowledge

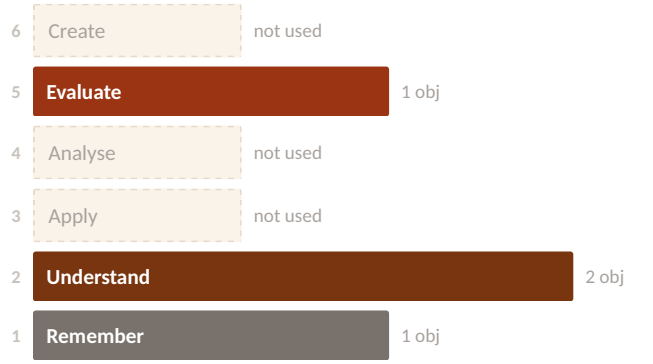
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**5**	—	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Describe reference, ingredients, therapeutic properties dosage and all administration details alongwith Anupana of each formulation.	*describe*	Conceptual ◦	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Explain Pathya Apathya, Sevana Avadhi (duration of treatment) and shelf-life of each formulation.	*explain*	Conceptual ◦	Must know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Interpret probable mode of action of each formulation as per Ayurveda.	*interpret*	Conceptual °	Desirable	Teach at **level 6**. The verb *interpret* is what the exam will ask for.
LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Explain research updates and clinical evidences of Arogyavardhini Gutika and Gandhaka Rasayana.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Critically analyse compound drugs for their therapeutic actions mentioned in the classics.	*critically*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	1	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Evaluation* — decide something, write down the reasons you used, then argue the opposite side.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PAPER 2 · UNIT 5

Aushadhi Kalpa-II (Compound Drugs/Formulations)

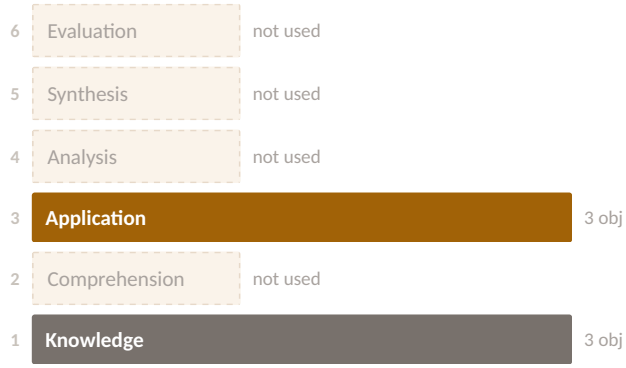
Hours: 14 lecture · 2 non-lecture | Term 3 | Theory marks: 15 | **7 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**1**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 2 nice to know	kind of knowledge

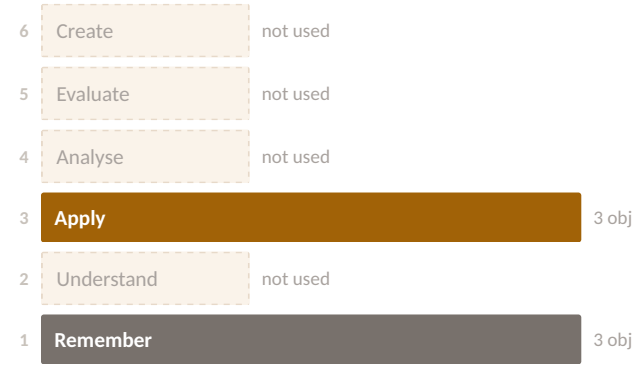
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**5**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Panchaguna Taila, Dadimavaleha, Bramhi Ghrita.	*describe*	Conceptual °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Describe ingredients, with its dose, therapeutic importance anupana and pathya-apathya along with its clinical evidences and Research updates.	*describe*	Conceptual	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist the formulations with its dose, anupana and indications.	*enlist*	Factual °	Nice to know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Describe Ingredients, therapeutic importance, dose, anupana pathya-pathya, duration of treatment, sevana kala, shelf life research updates and clinical evidences of Dashamoola Kwatha Pushyanuga Churna, Sanjivani Vati, Chitrakadi Gutika Simhanada Guggulu, Yogaraja Guggulu, Chyavanaprashavaleha Gandhakadya Malahara, Ashokarishta, Kutajarishta Panchagavya Ghrita, Bilvadi Gutika.	*describe*	Conceptual	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Mahamanjistadi kwatha, Sudarshana Churna Vyoshadi Vati, Bala Chaturbhadra Rasa, Lavana Bhaskara Churna, Narayana Taila,, Neeliringadi Taila Aravindasava Kumaryasava.	*describe*	Conceptual °	Nice to know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Understand & Recite following classical kalpas. Describe Amayika prayoga of each formulation. Understand probable mode of action as per AyurvedaSearch and. Record relevant reaserch articles. Create charts describing details of these drugs. Review market avaibality & poplularities of these drugs among practicing vaidyas & near by population.	*understand*	Factual °	Must know	Teach at **level 2**. The verb *understand* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Explain ethical aspect of administration of compound drugs mainly regarding duration of the treatment given.	*explain*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	3	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 7 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 2 · UNIT 6

Dosage Forms & Cosmetic Products

Hours: 5 lecture · 2 non-lecture | Term 3 | Theory marks: 5 | **12 objectives** | Highest level asked for: **Knowledge / Remember**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
11	**1**	**5**	**Conceptual**
levels 1–6	Krathwohl domain	6 desirable · 1 nice to know	kind of knowledge

WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3

6	Evaluation	not used
5	Synthesis	not used
4	Analysis	not used
3	Application	not used
2	Comprehension	not used
1	**Knowledge**	11 obj

REVISED 2001 — Anderson & Krathwohl

6	Create	not used
5	Evaluate	not used
4	Analyse	not used
3	Apply	not used
2	Understand	not used
1	**Remember**	11 obj

The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
3	**9**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 11 OBJECTIVES				
Define the term- Dosage forms.	*define*	Factual	Must know	Teach at this level.
Discuss the need of different dosage forms.	*discuss*	Conceptual ◦	Must know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Explain the classification of different dosage forms.	*explain*	Conceptual	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Enlist the solid / liquid / semisolid dosage forms.	*enlist*	Factual ◦	Must know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Discuss the routes of administration of different dosage forms.	*discuss*	Conceptual °	Must know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Describe the advantages and disadvantages of currently available dosage forms.	*describe*	Conceptual °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Discuss the research updates about modification of classical Ayurvedic dosage forms.	*discuss*	Conceptual °	Desirable	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Discuss the relevant case studies of different dosage forms.	*discuss*	Conceptual °	Nice to know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Define the term Cosmetics.	*define*	Factual	Desirable	Teach at this level.
Explain the classification of cosmetics based on their application on the specific body parts.	*explain*	Conceptual	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Differentiate between the ancient cosmetics and modern day cosmetics.	*differentiate*	Conceptual °	Desirable	Teach at **level 4**. The verb *differentiate* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
compare and appraise herbal cosmetics vs synthetic products.	*compare*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

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TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	11	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 11 of these 12 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
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FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days.
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PAPER 2 · UNIT 7

Nutraceuticals

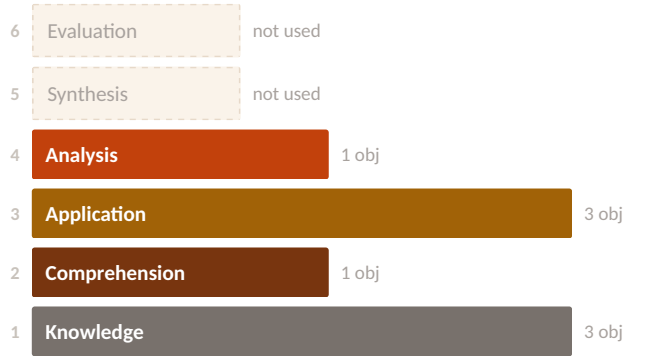
Hours: 6 lecture · 1 non-lecture | Term 3 | Theory marks: 5 | **9 objectives** | Highest level asked for: **Analysis / Analyse**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
8	**1**	**4**	**Conceptual**
levels 1–6	Krathwohl domain	4 desirable · 1 nice to know	kind of knowledge

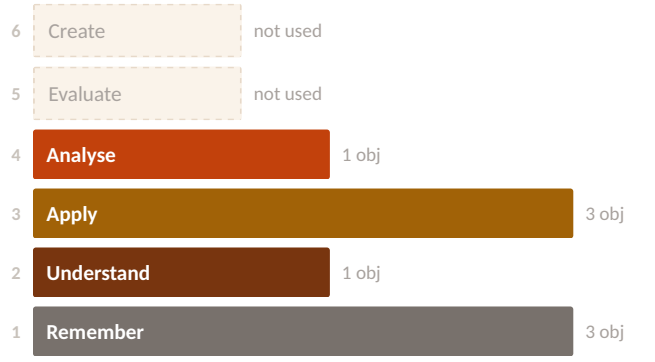
WHAT THIS TOPIC ASKS STUDENTS TO DO

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BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



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WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
3	**7**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Describe importance of combination of aahara and aushadha.	*describe*	Conceptual	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain Nutraceuticals with its types.	*explain*	Conceptual	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Explain research updates about nutraceuticals.	*explain*	Conceptual °	Nice to know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Enlist atleast two examples of Aushadhi siddha aahara with its indications for each category-->(Aharadravya belonging to following category and kalpa prepared from that aahara dravya and other aushadhi ingredients)1) Jala varga2) Dugdha varga3) Ikshu varga4) Suka-dhanya varga5) Shimbi Dhanya Varga6) Shaka Varga7) Phala Varga8) Lavana Varga9) Spices mhita.	*enlist*	Factual °	Desirable	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Describe with examples mode of action of Ayurvedic nutraceuticals mhita.	*describe*	Conceptual °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist at least two dietary preparations from Ayurvedic classics which can serve as rasayana/ nutraceutical in-1) General Health2) Pediatric Health3) Geriatric Health4) Women(Garbhini/ Sutika) Health5) Cardio-protection6) Chronic illness-recovery stage.	*enlist*	Factual °	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Describe potential of Ayurvedic diet and Rasayana in sports endeavor.	*describe*	Conceptual °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 4 · ANALYSIS / ANALYSE — 1 OBJECTIVES				
Corelate rasayana and nutraceutical with examples.	*correlate*	Conceptual °	Desirable	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
creat list of unique features of Ayurvedic neutraceuticals and appraise their current need.	*create*	Factual + Conceptual °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	3	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	1	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 4 of these 9 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 4, repeating it does not mean you know it.

PAPER 2 · UNIT 8

Anupana Prayoga for Aushadhi Kalpa

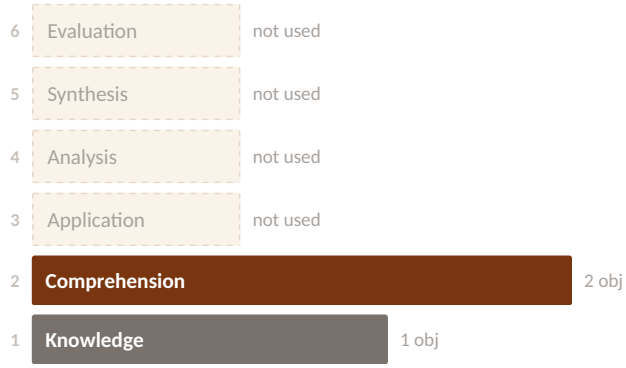
Hours: 4 lecture · 1 non-lecture | Term 3 | Theory marks: 5 | **4 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
3	**1**	**3**	**Factual**
levels 1–6	Krathwohl domain	1 desirable · 0 nice to know	kind of knowledge

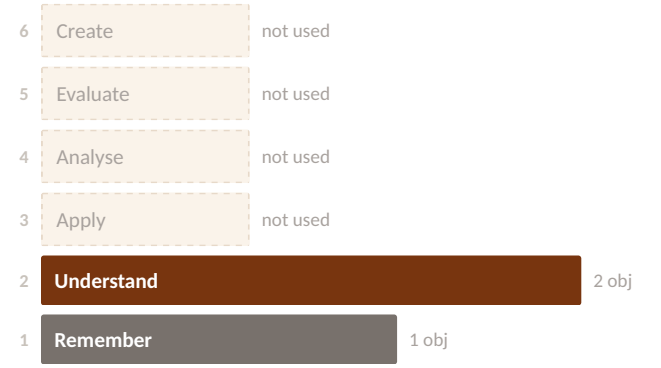
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**2**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Define and classify Anupana and Sahapana.	*define*	Factual ◦	Must know	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Explain different factors to be considered for selection of Anupana as per Disease and Patients.	*explain*	Conceptual ◦	Must know	Teach at this level.
Enlist different anupana for a single drug based on the condition of patient and disease. Elaborate with the help of examples of kalpas.	*enlist*	Factual ◦	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Describe value of proper selection of Anupanas with examples.	*describe*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 4 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Factual* — recall it until it sticks; *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 2 · UNIT 9

Aushadhi Prayoga Marga

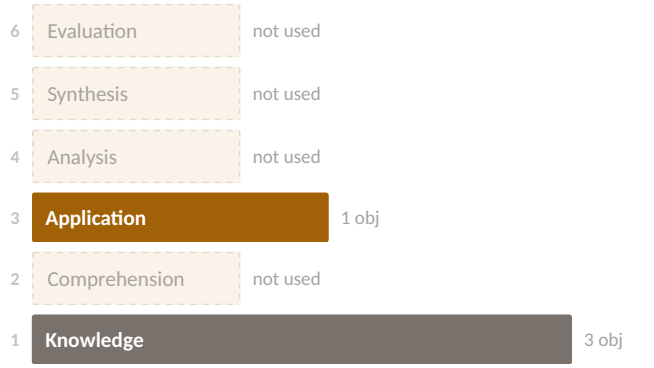
Hours: 5 lecture · 1 non-lecture | Term 3 | Theory marks: 10 | **5 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
4	**1**	**2**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 1 nice to know	kind of knowledge

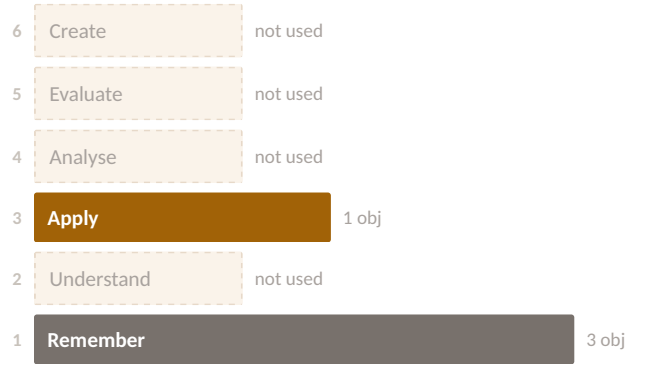
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**4**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Discuss the various types of Aushadhi Prayog Marga (route of drug administration).	*discuss*	Conceptual	Must know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Describe the advantages and disadvantages of each Aushadhi Prayog Marga.	*describe*	Conceptual °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist the dosage forms used in the specific Aushadhi Prayog Marga.	*enlist*	Factual °	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Discuss the nature of drug (s) administered in various routes of drug administration.	*discuss*	Conceptual °	Nice to know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Appraise administration of drugs through various routes mentioned in Ayurvedic treatment.	*appraise*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 2 · UNIT 10

Rational prescription along with safe dispensing of Ayurvedic formulations.

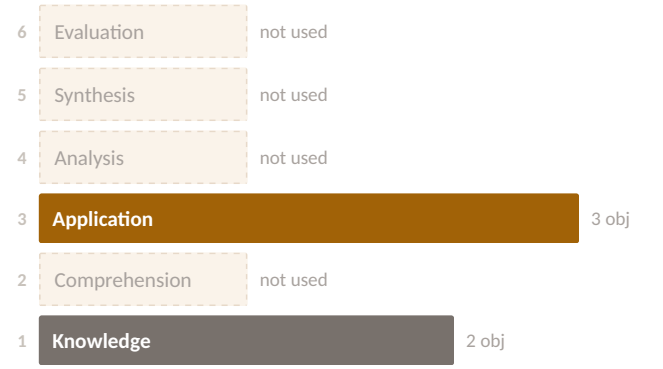
Hours: 1 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
5	**1**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	3 desirable · 0 nice to know	kind of knowledge

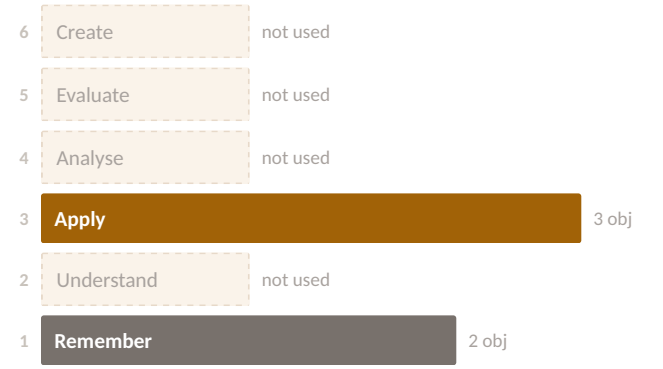
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**4**	**2**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Explain the importance of rational prescribing of drugs and the concept of essential drugs.	*explain*	Conceptual	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Describe the standard protocol for safe dispensing of Ayurvedic drugs.	*describe*	Procedural °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Describe and write demo ideal prescription.	*describe*	Conceptual °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain the safe dispensing and efficacious use of Ayurvedic drugs.	*explain*	Conceptual °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Demonstrate and educate home remedies to small group of population.	*demonstrate*	Procedural °	Desirable	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
explain ethical aspects related to prescription writing.	*explain*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	3	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 2 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 2 · UNIT 11

Traditional & Local health Practices

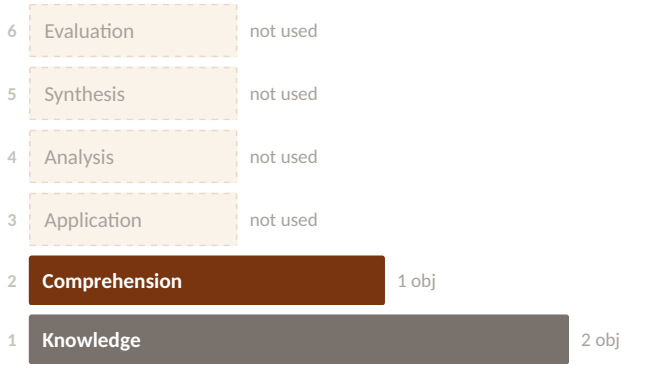
Hours: 2 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **4 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
3	**1**	**0**	**Factual**
levels 1–6	Krathwohl domain	4 desirable · 0 nice to know	kind of knowledge

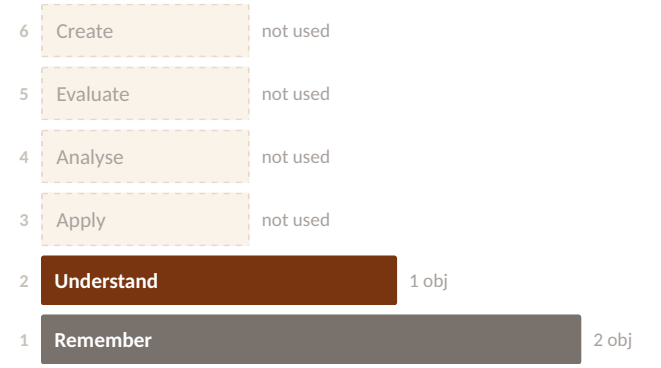
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
3	1	—	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Recognise the possible potential of product development and research based on Traditional knowledge.	*recognise*	Factual ◦	Desirable	Teach at this level.
Identify the factors responsible for grant of patent and erroneous grant of patent on indian traditional knowledge.?	*identify*	Factual ◦	Desirable	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Identify Local Health Traditions and Healing Knowledge.	*identify*	Factual ◦	Desirable	Teach at **level 1**. The verb *identify* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Appraise traditional knowledge of Ayurvedic medicines.	*appraise*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 4 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Factual* — recall it until it sticks; *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 2 · UNIT 12

Pharmacovigilance for Ayurveda drugs

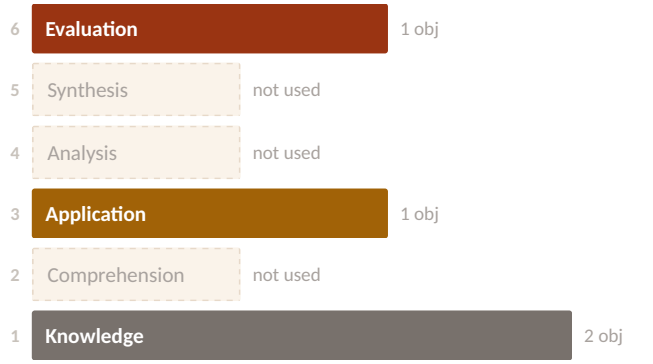
Hours: 1 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
4	**2**	**1**	**Conceptual**
levels 1–6	Krathwohl domain	5 desirable · 0 nice to know	kind of knowledge

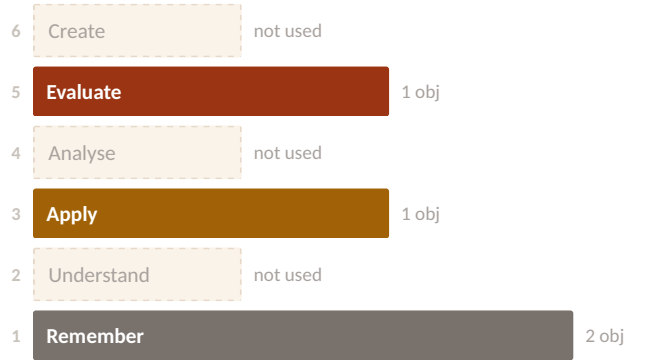
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**5**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Describe the term Pharmacovigilance and explain importance of Pharmacovigilance for Ayurvedic drugs.	*describe*	Factual + Conceptual ◦	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain the status and central sector scheme of Pharmacovigilance for Ayurveda, Siddha, Unani, and Homeopathy (ASU & H) drugs.	*explain*	Conceptual ◦	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Discuss and make critical comments on the safe and efficacious use of Ayurvedic drugs.	*discuss*	Conceptual °	Desirable	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Define Adverse Drug Reactions (ADR) and its types QZ, TT-Theor.	*define*	Conceptual °	Desirable	Teach at **level 1**. The verb *define* is what the exam will ask for.
AFFECTIVE DOMAIN — 2 OBJECTIVES				
Identify and monitor ADRs.	*identify*	Factual °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Debate on ADR of ASU drugs.	*debate*	Conceptual °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	1	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 2 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
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- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it; *Evaluation* — decide something, write down the reasons you used, then argue the opposite side.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PRACTICAL · UNIT 1

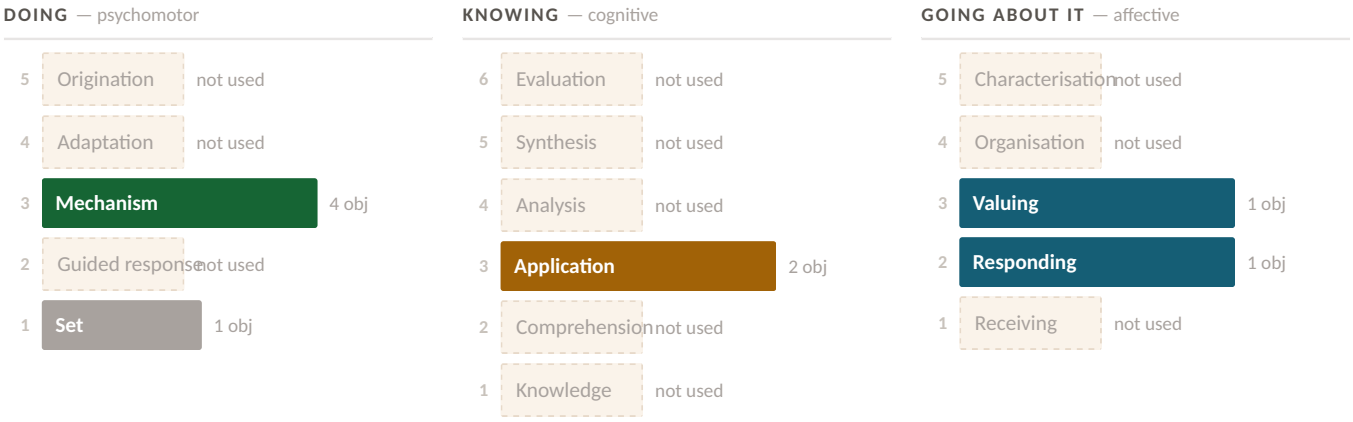
Paribhasha concept based Practicals

NCISM Table 4 | **9 objectives** | Psychomotor 5 · Cognitive 2 · Affective 2 | Highest doing level: **Mechanism**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
5	**2**	**2**	**6**
what they can do	what they must know	how they work	3 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 1 · SET — 1 OBJECTIVES				
Identify the drugs and recite classical name, English name and chemical composition and varga(class). Recite Maharasa uparasa, Sadharana rasa shlokas from Rasaratna samuchchaya.	*identify*	Factual	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
DOING — LEVEL 3 · MECHANISM — 4 OBJECTIVES				
Demonstrate the correct procedure / SOP and assess the changes observed after the procedure. Discuss the relevant samskara and its role in that procedure. Interpret the Physical, Chemical and Biological alterations.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Observe and describe the pharmaceutical preparation of Gandhaka druti. Explain the term alotropism. Recite melting boiling and evaporating temperature of Gandhaka.	*observe*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. They work unaided. The verb *observe* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
Demonstrate preparation of Kajjali. Recite its classical shloka. Prepare a checklist of parameters for its siddhilakshana. Interpret law of definite proportion to explain and calculate amount of free sulfur in the prepared amount of kajjali. Calculate the expected weight of rasasindura from the given amount of kajjali.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Identify relevant Bhasmapariksha and demonstrate samanya and vishesha Bhasma pariksha. Interpret physical/ chemical laws relevant to classical bhasma pariksha.	*identify*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 2 OBJECTIVES				
Observe the procedure of Vanga Jarana. Explain the difference between Jarana and Jaranaa. Discuss chemical process of oxidation and reduction. Recall the information about melting points of all the metals.	*observe*	Procedural	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
Identify various weights and recite essential measures from conversion chart of AFI. Recognise weighing machines and their weighing capacity.	*identify*	Factual	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
GOING ABOUT IT — LEVEL 2 · RESPONDING — 1 OBJECTIVES				
Appraise ancient indications mentioned in classics for confirmation of end point of the pharmaceutical process(Siddhi lakshana).	*appraise*	Conceptual + Procedural °	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
Identify and value SOP of each procedure of the practical conducted.	*identify*	Procedural	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	1	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	4	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	—	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	5	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Checklist CHK	Observed	2	Shows performance directly. The only kind of evidence that settles a doing objective.
P-EXAM P-EXAM	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Identification P-ID	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
Practical performance P-PRF	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
P-REC P-REC	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
PRN PRN	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

This unit has **5 doing objectives** but observed tools are set **4 times**. Spoken tools are set 5 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 2

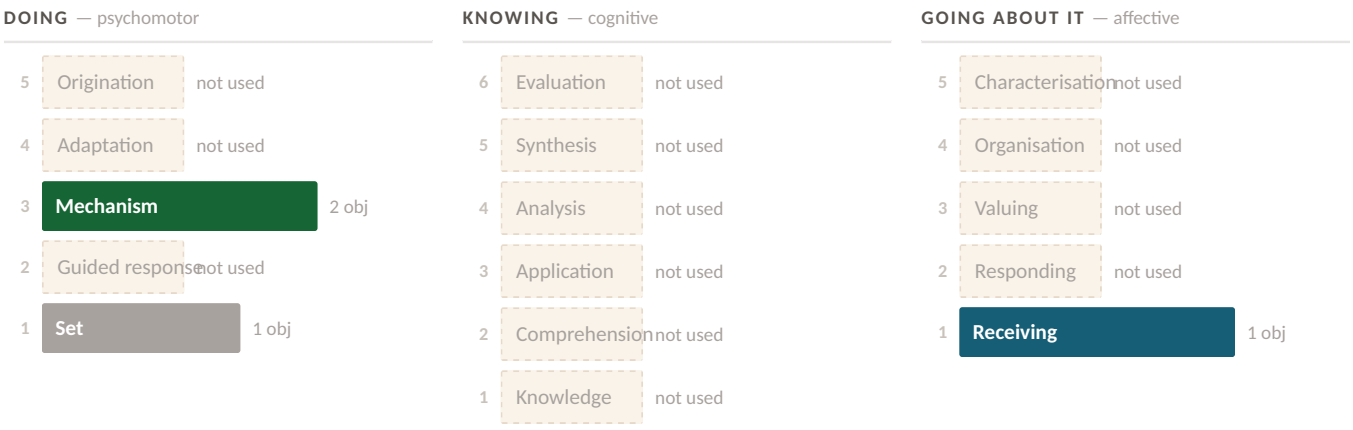
Panchavidha Kashaya Kalpana & their Upakalpana Practicals

NCISM Table 4 | **4 objectives** | Psychomotor 3 · Cognitive 0 · Affective 1 | Highest doing level: **Mechanism**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
3	**0**	**1**	**4**
what they can do	what they must know	how they work	0 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 1 · SET — 1 OBJECTIVES				
Recall, Identify and authenticate the raw materials required as per the dosage form/ formulations.	*recall*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching. The verb *recall* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
DOING — LEVEL 3 · MECHANISM — 2 OBJECTIVES				
Demonstrate the method of preparation to get the desired dosage form following SOP.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Assess and explain different parameters to achieve desired characters/ end points (siddhi lakshana) as per classical and contemporary parameters.	*assess*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
GOING ABOUT IT — LEVEL 1 · RECEIVING — 1 OBJECTIVES				
Describe ethical responsibility expected during Pharmaceutical preparation of Ayurvedic Formulations.	*describe*	Conceptual	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	1	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	2	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	—	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	3	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Practical performance P-PRF	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.

This unit has **3 doing objectives** but observed tools are set **1 time**. Spoken tools are set 3 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 3

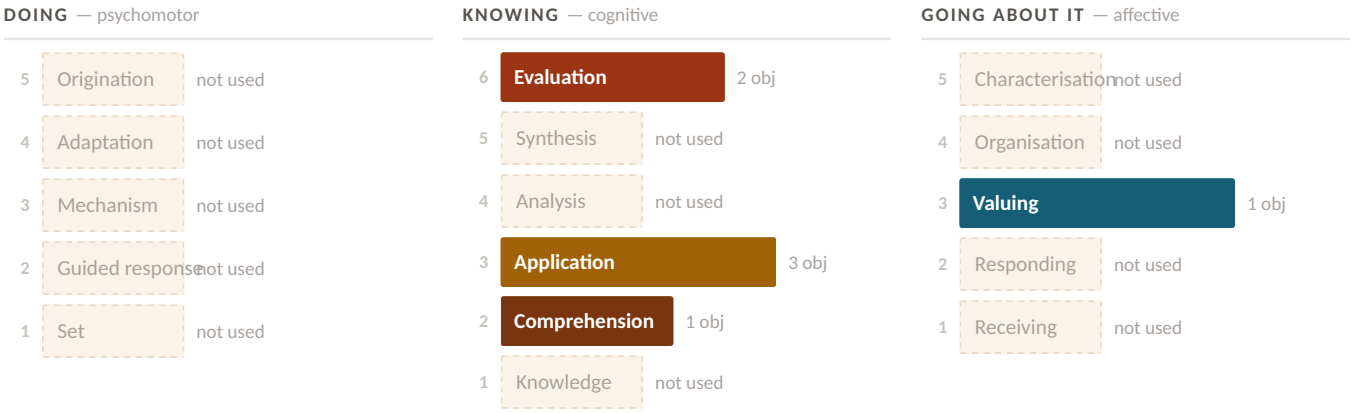
Rasa Dravya Aushadhi Nirmana Practicals

NCISM Table 4 | **7 objectives** | Psychomotor 0 · Cognitive 6 · Affective 1 | Highest doing level: **none — no psychomotor objective here**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
0	**6**	**1**	**6**
what they can do	what they must know	how they work	1 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Explain reference Shloka of kalpas & write with interpretation -list of ingredients, their proportion, principles of yoga - yoga samyojana dravya.	*explain*	Factual + Conceptual °	Must know	Spoken or written answer is fair here.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Identify raw Dravya, used part with classical Mana and interpret in metric system.	*identify*	Factual °	Must know	Spoken or written answer is fair here.
Identify, Utilize & know mechanism of Yantra used for drug preparation.	*identify*	Procedural °	Must know	Spoken or written answer is fair here.
Demonstrate Sidhdhi Lakshna.	*demonstrate*	Procedural °	Must know	Spoken or written answer is fair here.
KNOWING — LEVEL 6 · EVALUATION / EVALUATE — 2 OBJECTIVES				
Observe & Record various parameters responsible for good manufacturing which are done during process of pharmaceutical preparation.	*observe*	Procedural	Must know	Spoken or written answer is fair here.
Assess finished product as per classics.	*assess*	Conceptual °	Must know	Spoken or written answer is fair here.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
Appraise peculiarities of chaturvidha rasayana preparation.	*appraise*	Conceptual	Desirable	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	3	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-EXAM P-EXAM	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Viva VV-Viva	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Poster M-POS	Product	1	Shows the result. Say in advance what you will mark it against.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 4

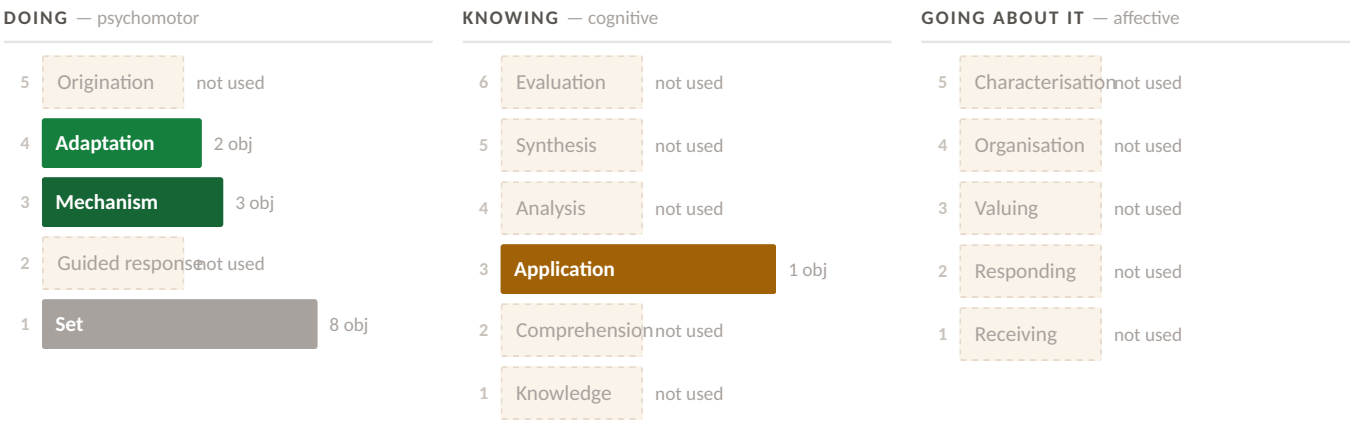
Bheshaja Kalpana Practicals -I

NCISM Table 4 | **14 objectives** | Psychomotor 13 · Cognitive 1 · Affective 0 | Highest doing level: **Adaptation**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
13	**1**	**0**	**6**
what they can do	what they must know	how they work	5 desirable · 3 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 1 · SET — 8 OBJECTIVES				
Demonstrate Arka preparation method, along with its packaging technique and therapeutic importance.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
Identify the genuine sample of Guggulu and demonstrate preparation of different types of Guggulu along with drying techniques.	*identify*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
Identify the useful part of the raw material and demonstrate Sattva preparation method.	*identify*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
Demonstrate preparation of Varti along with drying and packaging techniques.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
Demonstrate Lavana preparation methods, along with its packaging techniques.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. You may guide while watching.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
Demonstrate preparations of Sikta taila along with identify the genuine sample of Sikta. Perform and. Describe Malahara kalpana preparation method with its confirmatory tests of end point.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
Demonstrate preparation of different types of Malahara with its applied aspect.	*demonstrate*	Procedural	Nice to know	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
Demonstrate preparation of different types of Lepa and preparation of Shatadhouta ghrita.	*demonstrate*	Procedural	Nice to know	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 3 OBJECTIVES				
Demonstrate preparation of Vati along with drying techniques.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate preparation and application aspects of Upanaha kalpana.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. They work unaided.
Perform and. Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.	*perform*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. They work unaided.
DOING — LEVEL 4 · ADAPTATION — 2 OBJECTIVES				
Identify the internal and external applications of different kshara kalpana along with importance of ksharasutra through surgical aspects.	*identify*	Conceptual + Procedural °	Must know	**Watch them do it.** Checklist or OSPE station. Alter a condition and watch the adjustment. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
Demonstrate and. Explain preparation methods of Masi kalpana for its internal or external application in clinical practice.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. Alter a condition and watch the adjustment.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Observe demonstration of formulation of Danta manjana.	*observe*	Factual °	Nice to know	Spoken or written answer is fair here, but pair it with the doing objective it supports.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	8	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	3	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	2	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	9	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Viva, knowledge check V-KC	Spoken	4	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
PRN PRN	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-EXAM P-EXAM	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

This unit has **13 doing objectives** but observed tools are set **0 times**. Spoken tools are set 13 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 5

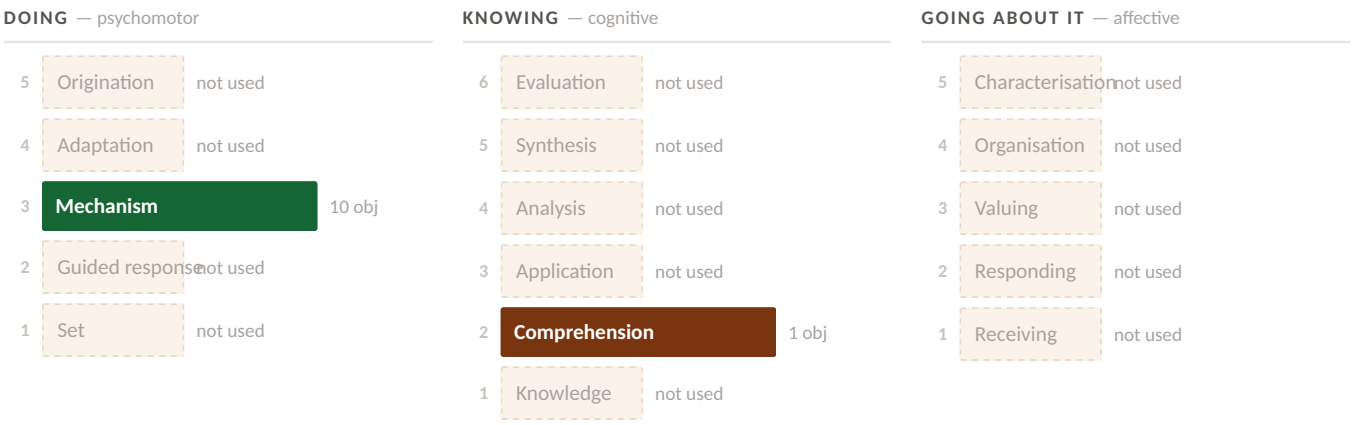
Bheshaja Kalpana Practicals -II

NCISM Table 4 | **11 objectives** | Psychomotor 10 · Cognitive 1 · Affective 0 | Highest doing level: **Mechanism**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
10	**1**	**0**	**7**
what they can do	what they must know	how they work	2 desirable · 2 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 3 · MECHANISM — 10 OBJECTIVES				
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Ghrita Murchana and Taila Murchana.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Jatyadi Gritha/ Triphala Gritha/ Ksheera Shatphala Gritha/Phala Gritha.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Ksheera Bala Taila/Kasisadi Taila/ Panchaguna Taila/ Arka Taila/Kutajasuryapaki taila.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Bhallataka Taila Patana/ Jayapala Taila Patana/Vishvamitra Kalpa Sneha.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. They work unaided.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Vasavaleha/ Chavana Prasha Avaleha/ Kushmanda Avaleha.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Nimbu Sharkara.	*demonstrate*	Procedural	Nice to know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Daruharidra Rasakriya.	*demonstrate*	Procedural	Nice to know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Haridra Khanda/ Narikela Khanada.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kumaryasava/Drakshasava.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kutajarista/Ashokarishta/Takrarishta.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlist the results, packing techniques and storage of unique formulations for e.g. Kanji/ Madushukta.	*demonstrate*	Procedural	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	—	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	10	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	—	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
OSPE station OSPE	Observed	6	Shows performance directly. The only kind of evidence that settles a doing objective.
P-EXAM P-EXAM	Spoken	6	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Practical viva P-VIVA	Spoken	5	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-REC P-REC	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
PRN PRN	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Compilation COM	Record over time	1	Shows habit and consistency. The right tool for how a student goes about the work.

This unit has **10 doing objectives** but observed tools are set **6 times**. Spoken tools are set 5 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
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- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 6

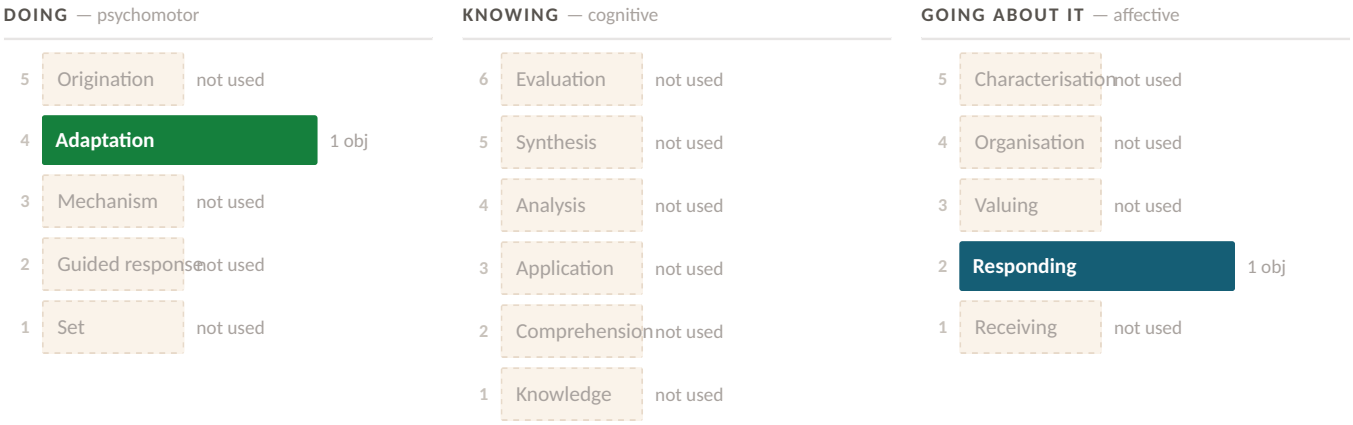
Dosage Forms & Self-care Products Practicals

NCISM Table 4 | **2 objectives** | Psychomotor 1 · Cognitive 0 · Affective 1 | Highest doing level: **Adaptation**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
1	**0**	**1**	**0**
what they can do	what they must know	how they work	2 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 4 · ADAPTATION — 1 OBJECTIVES				
Observe instruments used to prepare with one example for each their method of preparation, ingredients used with their quantityand Quality control Parameters.of following self care productsHair care: Shampoo Body care: Soap, Perfume Face care: Face Pack/Talcum Powder Lip Care: Lip Balm, Lipstick Oral care: Tooth Paste/ Mouth Wash Foot Care: Foot Cream Hand Disinfectant: Hand SanitizerSkin Care: Moisturizer, Sunscreen Lotion following self care products.	*observe*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. Alter a condition and watch the adjustment. The verb *observe* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
GOING ABOUT IT — LEVEL 2 · RESPONDING — 1 OBJECTIVES				
Observe instruments used to prepare soliddosage form, liquid dosage form and Semisolid Dosage Form with one example for each, their method of preparation, ingredients used with their quantity and Quality control Parameters.	*observe*	Procedural	Desirable	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	—	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	—	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	1	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Checklist CHK	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
P-EXAM P-EXAM	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 7

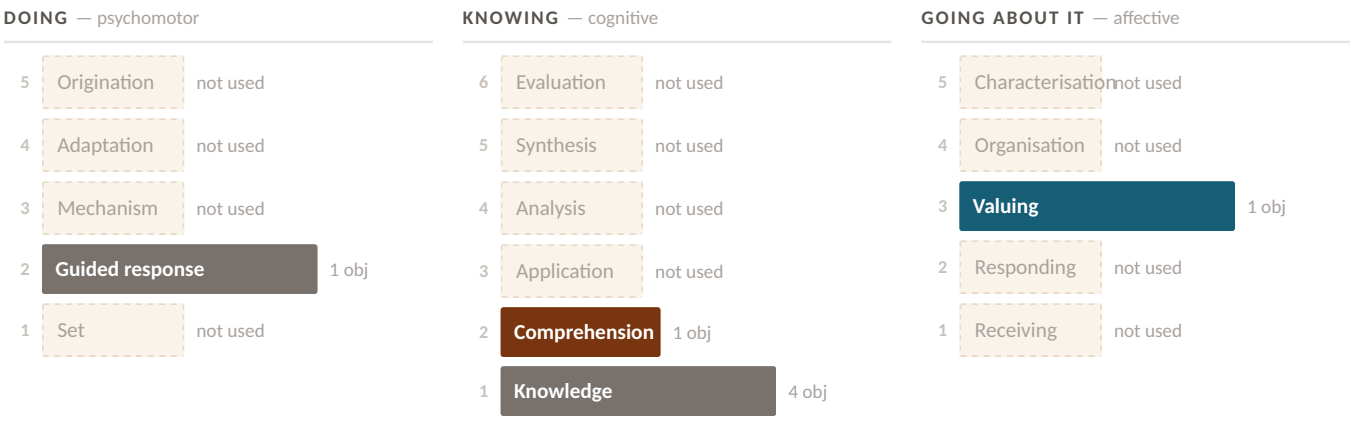
Field Visit/ Study Tour

NCISM Table 4 | **7 objectives** | Psychomotor 1 · Cognitive 5 · Affective 1 | Highest doing level: **Guided response**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
1	**5**	**1**	**0**
what they can do	what they must know	how they work	6 desirable · 1 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 2 · GUIDED RESPONSE — 1 OBJECTIVES				
Prepare own products as per prevalent FDA guidelines.	*prepare*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
KNOWING — LEVEL 1 · KNOWLEDGE / REMEMBER — 4 OBJECTIVES				
Field Visit - Record the storage condition of the raw / in-process / finished goods prepared in the approved sections.	*field*	Procedural	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
Determine the role of various Yantras / machineries used in the approved sections of the Pharmacy.	*determine*	Conceptual	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
Enlist the documents required in Batch Manufacturing of Ayurvedic medicines.	*enlist*	Factual °	Nice to know	Spoken or written answer is fair here, but pair it with the doing objective it supports.
Identify the role of Quality Control instruments and equipments in ensuring a quality Ayurvedic product.	*identify*	Conceptual °	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Differentiate between the ancient and contemporary methods of drug preparation.	*differentiate*	Procedural	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
Appreciate the importance of Good Manufacturing Practices and Good Packaging Practices required in Ayurvedic Drug manufacturing.	*appreciate*	Conceptual	Desirable	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	—	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	1	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	—	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	—	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
PRN PRN	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
DEB DEB	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Compilation COM	Record over time	1	Shows habit and consistency. The right tool for how a student goes about the work.
Poster M-POS	Product	1	Shows the result. Say in advance what you will mark it against.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-EXAM P-EXAM	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

This unit has **1 doing objectives** but observed tools are set **0 times**. Spoken tools are set 0 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 8

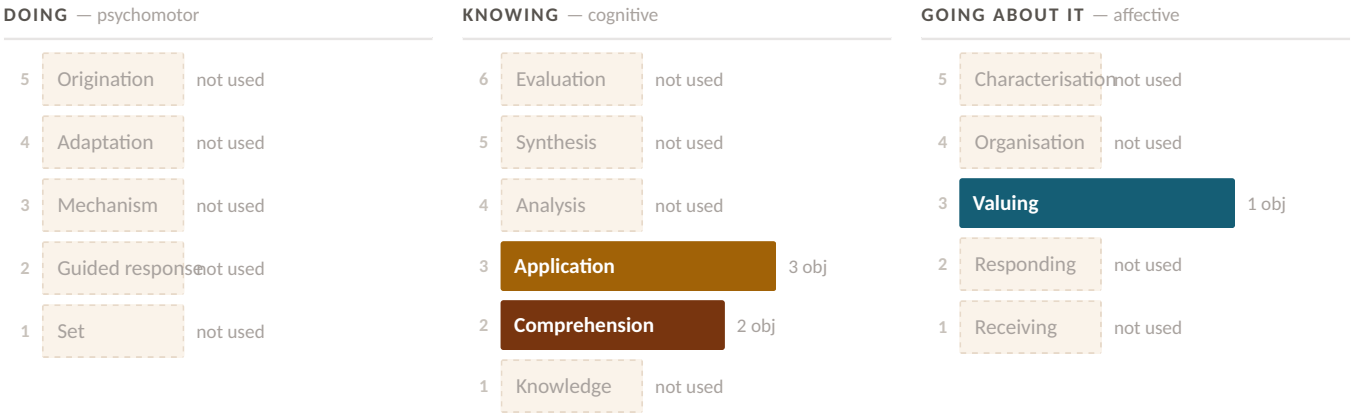
Hospital IPD Practical

NCISM Table 4 | **6 objectives** | Psychomotor 0 · Cognitive 5 · Affective 1 | Highest doing level: **none — no psychomotor objective here**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
0	**5**	**1**	**2**
what they can do	what they must know	how they work	3 desirable · 1 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Review & Observe the case Sheets.	*review*	Conceptual °	Nice to know	Spoken or written answer is fair here.
Discuss for formulation prescription manner.	*discuss*	Conceptual °	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Demonstrate & Explain prescription method.	*demonstrate*	Procedural	Desirable	Spoken or written answer is fair here.
Measure dose of prescribed drug.	*measure*	Procedural °	Must know	Spoken or written answer is fair here.
Enlist Time, Duration, Pathya, Apathya while prescribing drugs.	*enlist*	Factual °	Desirable	Spoken or written answer is fair here.
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
value details and methodical writing of case records of hospital IPD patients.	*value*	Conceptual	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Viva, knowledge check V-KC	Spoken	3	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Case study T-CS	Written	1	Shows reasoning. Pair it with observation where the objective is about doing.
P-PS P-PS	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Practical performance P-PRF	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 9

Drug Dispensing Practical

NCISM Table 4 | **4 objectives** | Psychomotor 0 · Cognitive 4 · Affective 0 | Highest doing level: **none — no psychomotor objective here**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
0	4	0	0
what they can do	what they must know	how they work	3 desirable · 1 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.

DOING — psychomotor

5	Origination	not used
4	Adaptation	not used
3	Mechanism	not used
2	Guided response	not used
1	Set	not used

KNOWING — cognitive

6	**Evaluation**	1 obj
5	Synthesis	not used
4	Analysis	not used
3	**Application**	2 obj
2	**Comprehension**	1 obj
1	Knowledge	not used

GOING ABOUT IT — affective

5	Characterisation	not used
4	Organisation	not used
3	Valuing	not used
2	Responding	not used
1	Receiving	not used

The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Determine and identify the cause of common errors occurred during dispensing of Ayurvedic drugs.	*determine*	Conceptual °	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 2 OBJECTIVES				
Assess the arrangement of Ayurvedic drugs according to nature and type in dispensing room or pharmacy.	*assess*	Conceptual °	Nice to know	Spoken or written answer is fair here.
Explain use of correct drug supply to the right patients, in the required dosage, quantities and clear drug information.	*explain*	Conceptual °	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Assess practice of prescription processing and labelling of the drugs.	*assess*	Conceptual °	Desirable	Spoken or written answer is fair here.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Viva, knowledge check V-KC	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Practical performance P-PRF	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
PRN PRN	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 10

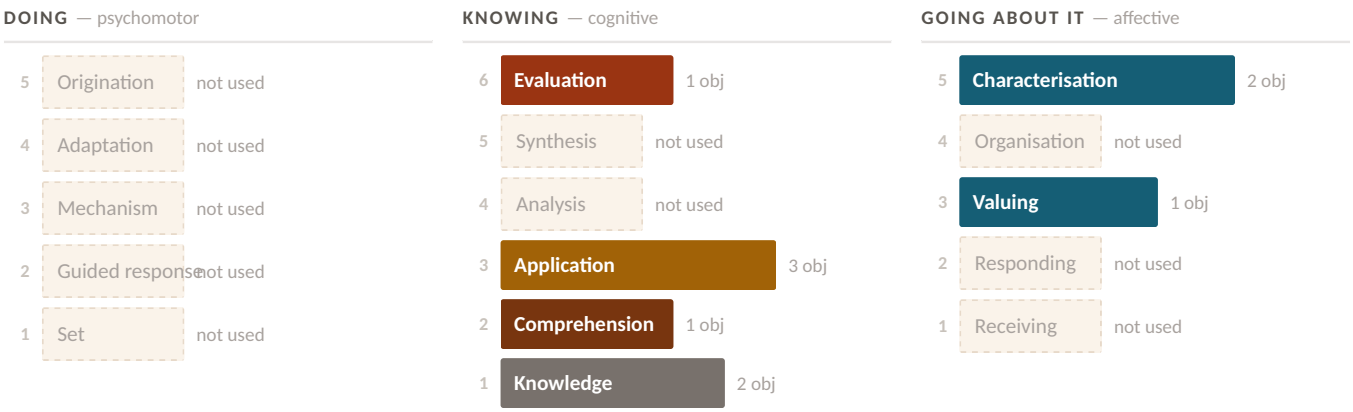
Quality Control Practicals

NCISM Table 4 | **10 objectives** | Psychomotor 0 · Cognitive 7 · Affective 3 | Highest doing level: **none — no psychomotor objective here**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
0	7	3	7
what they can do	what they must know	how they work	3 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Develop analytical skills for understanding Identity, Purity and Strength of raw materials and finished products as per the standard guidelines of the Ayurvedic Pharmacopoeia of India.	*develop*	Procedural °	Must know	Spoken or written answer is fair here.
Identify the geological description of minerals: physical parameters.	*identify*	Procedural °	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Demonstrate the calibration techniques used for Weighing Balance (chemical and physical), pH Meter, Hot Air Oven and Electric Muffle Furnace to ensure the accuracy of the instrument what it is intended to measure.	*demonstrate*	Procedural	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Describe ayurvedic perspectives of quality control parameters applying for solid, semisolid and liquid dosage forms.	*describe*	Procedural °	Must know	Spoken or written answer is fair here.
Demonstrate bulk density, loss on drying, total ash, pH and water soluble extractives of Churna A book.	*demonstrate*	Procedural °	Must know	Spoken or written answer is fair here.
Determine iodine value, acid value and saponification value of oils / ghee.	*determine*	Conceptual	Must know	Spoken or written answer is fair here.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Calculate the dose of various dosage forms and their dispensing methods.	*calculate*	Procedural	Must know	Spoken or written answer is fair here.
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
Appraise quality control procedures done for Ayurvedic formulations.	*appraise*	Conceptual + Procedural °	Desirable	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.
GOING ABOUT IT — LEVEL 5 · CHARACTERISATION — 2 OBJECTIVES				
Perform the analytical tests of tablets/ vati/ gutika for hardness uniformity of weight and friability.	*perform*	Conceptual °	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.
Estimate specific gravity, alcohol content and total solids of Asava & Arishta.	*estimate*	Conceptual °	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	4	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-EXAM P-EXAM	Spoken	3	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Identification P-ID	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
PRN PRN	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
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- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PART C

Reference

Nine appendices. Everything here is worked out from the curriculum's own objectives, so the counts in these tables and the counts in the Units are the same numbers. Use C1 and C2 to decode anything in the syllabus document; C3 to compare Units at a glance when planning a term or setting a paper; C4 and C5 when writing or rewording objectives of your own.

1. **C1** The sixteen curriculum codes, decoded
2. **C2** The two schemes side by side
3. **C3** Level profile of every Unit
4. **C4** Verb index — every verb the curriculum uses, and its level
5. **C5** Kinds of knowledge across the curriculum
6. **C6** Level against priority
7. **C7** Teaching methods named in the curriculum
8. **C8** Assessment tools named in the curriculum
9. **C9** Sources and further reading

↑ Contents

PART C · APPENDIX C1

The sixteen curriculum codes, decoded

Every code that appears in Column C3 of Table 3 and Column C4 of Table 4, with the domain it belongs to, the level it sits at, and how many objectives carry it. This is the table to keep beside the syllabus document.

CODE	DOMAIN	LEVEL	NAME	TIMES USED IN AYUG-RB
CK	Thinking	Level 1	Knowledge / Remember	86
CC	Thinking	Level 2	Comprehension / Understand	68
CAP	Thinking	Level 3	Application / Apply	35
CAN	Thinking	Level 4	Analysis / Analyse	4
CS	Thinking	Level 5	Synthesis / Create	not used
CE	Thinking	Level 6	Evaluation / Evaluate	10
PSY-SET	Doing	Level 1	Set	10
PSY-GUD	Doing	Level 2	Guided response	1
PSY-MEC	Doing	Level 3	Mechanism	19
PSY-ADT	Doing	Level 4	Adaptation	3
PSY-ORG	Doing	Level 5	Origination	not used
AFT-REC	Going about it	Level 1	Receiving	1
AFT-RES	Going about it	Level 2	Responding	5
AFT-VAL	Going about it	Level 3	Valuing	21
AFT-SET	Going about it	Level 4	Organisation	1
AFT-CHR	Going about it	Level 5	Characterisation	4

Sixteen codes are defined; fourteen are used. **CS** (Synthesis / Create) and **PSY-ORG** (Origination) appear in the key but against no objective.

PART C · APPENDIX C2

The two schemes side by side

The six thinking levels in both schemes, with what a student actually does at each. Where a colleague speaks of "the top level", this table settles which one is meant.

LEVEL	BLOOM 1956, AS PRINTED IN THE CURRICULUM	REVISED 2001, ANDERSON & KRATHWOHL	WHAT THE STUDENT DOES
Level 1	**Knowledge**	**Remember**	recalls it
Level 2	**Comprehension**	**Understand**	restates it in their own arrangement
Level 3	**Application**	**Apply**	uses it on something new
Level 4	**Analysis**	**Analyse**	takes it apart and shows how the parts relate
Level 5	**Synthesis**	**Evaluate** ← the schemes differ here	makes something that was not given
Level 6	**Evaluation**	**Create** ← and here	judges it against stated criteria

The top two levels are the only place the two schemes disagree. In 1956, **Synthesis** is level 5 and **Evaluation** level 6. In 2001, **Evaluate** is level 5 and **Create** level 6. This book follows the 1956 numbering, because that is what the curriculum prints.

PART C · APPENDIX C3

Level profile of every Unit

All thirty-six Units in one view. This is the fastest way to see where a term's weight sits, and to check that the marks on a paper are going where the thinking actually is.

UNIT	TOPIC	OBJ	MK	LEVELS USED	HIGHEST	MUST KNOW
PAPER 1 — AYURVEDIYA AUSHADHI NIRMANA VIGYANA						
1	Chronological development of Ayurvediya Aush	7	5	1 2	Comprehension	3 of 7
2	Paribhasha (Terminology)	19	10	1 2 3 6	Evaluation	12 of 19
3	Adharbhuta Siddhanta (Application of fundame	11	5	1 2	Comprehension	9 of 11
4	Yantropakaranani- I (Equipments and machiner	6	5	1 2 3	Application	1 of 6
5	Yantropakaranani -II (Equipments, fuel and H	21	5	1 2 3	Application	3 of 21
6	Kalpana Nirmana I (Primary & Secondary dosag	11	10	1 2	Comprehension	7 of 11
7	Kalpana Nirmana-II (Method of Preparation of	4	10	1 3	Application	4 of 4
8	Rasa Dravya Parichaya- I	9	10	1 2	Comprehension	7 of 9
9	Rasa Dravya Parichaya II	7	5	1 2 6	Evaluation	1 of 7
10	Rasadravya Parichaya III	5	5	1 2	Comprehension	0 of 5
11	Kalpana Nirman -III (Method of Preparation o	6	10	2 3	Application	2 of 6
12	Chaturvidha Rasayana	6	10	2 3 4	Analysis	2 of 6
13	Current and emerging trend in Ayurvedic	6	5	1	Knowledge	0 of 6
14	GMP(Schedule T) & Regulatory aspects of Ayur	7	5	1 2	Comprehension	2 of 7
PAPER 2 — AYURVEDIYA AUSHADHI PRAYOGA VIGYANA						
1	Aushadhi Prayoga Vigyana	1	5	1	Knowledge	1 of 1
2	Single drug (Herbal & Mineral)	5	10	1 2	Comprehension	3 of 5
3	Single drug(Bhasma, Shuddha & Pishti)	5	15	1 2 3 6	Evaluation	2 of 5
4	Aushadhi Kalpa -I (Compound formulations)	5	15	1 2 6	Evaluation	2 of 5
5	Aushadhi Kalpa-II (Compound Drugs/ Formulatio	7	15	1 3	Application	3 of 7
6	Dosage Forms & Cosmetic Products	12	5	1	Knowledge	5 of 12
7	Nutraceuticals	9	5	1 2 3 4	Analysis	4 of 9
8	Anupana Prayoga for Aushadhi Kalpa	4	5	1 2	Comprehension	3 of 4
9	Aushadhi Prayoga Marga	5	10	1 3	Application	2 of 5
10	Rational prescription along with safe dispen	6	5	1 3	Application	3 of 6
11	Traditional & Local health Practices	4	5	1 2	Comprehension	0 of 4
12	Pharmacovigilance for Ayurveda drugs	6	5	1 3 6	Evaluation	1 of 6
PRACTICAL UNITS — NCISM TABLE 4						
1	1.Paribhasha concept based Practicals	9	—	1 3	Mechanism	6 of 9

UNIT	TOPIC	OBJ	MK	LEVELS USED	HIGHEST	MUST KNOW
2	2.Panchavidha Kashaya Kalpana & their Upakal	4	—	1 3	Mechanism	4 of 4
3	3.Rasa Dravya Aushadhi Nirmana Practicals	7	—	—	no doing objective	6 of 7
4	4.Bheshaja Kalpana Practicals -I	14	—	1 3 4	Adaptation	6 of 14
5	5.Bheshaja Kalpana Practicals -II	11	—	3	Mechanism	7 of 11
6	6.Dosage Forms & Self-care Products Practica	2	—	4	Adaptation	0 of 2
7	7.Field Visit/ Study Tour	7	—	2	Guided response	0 of 7
8	8.Hospital IPD Practical	6	—	—	no doing objective	2 of 6
9	9. Drug Dispensing Practical	4	—	—	no doing objective	0 of 4
10	10.Quality Control Practicals	10	—	—	no doing objective	7 of 10

For theory Units, **Levels used** and **Highest** refer to the thinking levels. For practical Units they refer to the doing levels, since that is what a practical is for. **Mk** is the theory marks the curriculum allots.

PART C · APPENDIX C4

Verb index — every verb the curriculum uses, and its level

The opening verb of all 268 objectives, sorted by the level that verb belongs to. The verb is what a student meets in the examination, which makes this the most practical table in the book.

LEVEL	NAME	VERBS THE CURRICULUM ACTUALLY USES, WITH HOW OFTEN
Level 1	**Knowledge** Remember	enlist 29, identify 17, define 11, observe 6, recognise 3, recall 2, recite 1
Level 2	**Comprehension** Understand	explain 49, describe 47, discuss 17, elaborate 2, illustrate 1, understand 1
Level 3	**Application** Apply	demonstrate 28, perform 2, prepare 1, measure 1, calculate 1
Level 4	**Analysis** Analyse	determine 7, choose 2, review 2, differentiate 2, categorize 1, compile 1, analyse 1, discriminate 1, compare 1, correlate 1
Level 5	**Synthesis** Create	develop 2, produce 1, create 1
Level 6	**Evaluation** Evaluate	assess 6, appraise 6, interpret 3, justify 2, critically 2
Affective	**Attitude verbs**	appreciate 3, debate 1, value 1

Use this when writing an objective: choose the level first, then take a verb from that row. Where an objective's verb sits at a different level from its code, the Unit in Part B says so and tells you which to pitch to.

PART C · APPENDIX C5

Kinds of knowledge across the curriculum

The second axis of the 2001 scheme, counted across all 268 objectives. An objective that works on two kinds is counted under both, which is why the totals here exceed 268.

KIND OF KNOWLEDGE	OBJECTIVES	HOW IT IS LEARNED, AND WHAT GOES WRONG WHEN IT IS TAUGHT AS SOMETHING ELSE
Factual	56	Rehearsal and self-testing: cover, retrieve, check, repeat across days. Taught as concept, students can discuss it but cannot produce the list under pressure.
Conceptual	147	Explaining it aloud to somebody without notes. Taught as fact, students can recite the principle and cannot use it on anything new.
Procedural	88	Carrying it out, more than once. Taught as fact or concept, this is the origin of "they know it but they cannot do it".
Metacognitive	not used	Being made to look back at how they worked. Not asked for anywhere in this curriculum.

OBJECTIVES WORKING ON MORE THAN ONE KIND

COMBINATION	OBJECTIVES	WHAT IT MEANS FOR TEACHING
Conceptual + Procedural	12	Part of the objective is learned one way and part another. Split the session accordingly rather than choosing one.
Factual + Conceptual	8	Part of the objective is learned one way and part another. Split the session accordingly rather than choosing one.
Factual + Procedural	1	Part of the objective is learned one way and part another. Split the session accordingly rather than choosing one.
Factual + Conceptual + Procedural	1	Part of the objective is learned one way and part another. Split the session accordingly rather than choosing one.

The kind of knowledge is read from the wording of each objective — its verb and what the verb acts on. Where the wording supports a reading without settling it, the Unit marks the label with a small circle (°).

PART C · APPENDIX C6

Level against priority

How Must know, Desirable and Nice to know are spread across the levels. Useful when deciding what can be moved to self-directed learning and what cannot.

LEVEL	NAME	MUST KNOW	DESIRABLE	NICE TO KNOW
Level 1	**Knowledge** / Remember	27	46	13
Level 2	**Comprehension** / Understand	35	26	7
Level 3	**Application** / Apply	21	10	4
Level 4	**Analysis** / Analyse	—	4	—
Level 5	**Synthesis** / Create	—	—	—
Level 6	**Evaluation** / Evaluate	4	6	—
Doing	all levels	20	9	4
Going about it	all levels	13	18	1

Priority is an independent judgement, not part of Bloom. It does not rise with level, and it should not be assumed to: an objective can be Must know at level 1 and Nice to know at level 6.

PART C · APPENDIX C7

Teaching methods named in the curriculum

The methods that appear in Column F3 and F4, grouped by the level each serves best. Expand the abbreviations here when reading the syllabus document.

BEST SUITED TO	METHODS THE CURRICULUM NAMES
Level 1	Lecture (L), Lecture with PowerPoint (L&PP), Lecture with video clips (L_VC), Recitation (REC), Game-based learning (GBL), Word puzzle (WP), Quiz (QZ), Puzzle (PUZ), Self-directed learning (SDL), Library session (LS)
Level 2	Discussion (DIS), Lecture and group discussion (L&GD), Flipped classroom (FC), Brainstorming (BS), Peer learning (PER), Peer learning (PL), Group interaction (GRI), Blended learning (BL), Mobile learning (ML)
Level 3 and above	Problem-based learning (PBL), Case-based learning (CBL), Project-based learning (PrBL), Team-based learning (TBL), Inquiry-based (IBL), Inquiry based (IB), Team project work (TPW), Practical training (PT), Demonstration (D), Demonstration with lecture (D_L), Simulation (SIM), Role play (RP), Field visit (FV), Real-life experience (RLE), Educational visit (EDU), Early clinical exposure (ECE), Debate (DEB), Presentation (PRN)

A method is not fixed to one level, but each has a level it serves best. Lecture builds level 1 well and level 2 poorly. From level 2 upward students must handle the material themselves, which is what the discussion and self-directed hours are for.

PART C · APPENDIX C8

Assessment tools named in the curriculum

The tools that appear in Column G3 and G4, grouped by what kind of evidence each produces. Match the kind to the objective before choosing the tool.

KIND	TOOLS THE CURRICULUM NAMES	WHAT THIS KIND CAN SHOW
Written	Theory test (TT-Theory, 16), Open book test (T-OBT, 10), Extended matching items (T-EMI, 5), Constructed response questions (T-CRQs, 3), Problem solving (P-PS, 2), Case study (T-CS, 2)	Reasoning, and thinking at any level if the stem is built for it.
Spoken	Practical viva (P-VIVA, 57), Viva, knowledge check (V-KC, 19), M-POSF (M-POSF, 9), QZ (QZ, 7), DEB (DEB, 7), Interview (C-INT, 2), M-CHTF (M-CHTF, 2), Viva (VV-Viva, 2), PUZ (PUZ, 1), WP (WP, 1), RK (RK, 1)	Whether a student knows and can explain. **It cannot show whether they can carry out a procedure.**
Observed	Practical examination (P-EXAM, 21), Checklist (CHK, 7), OSPE station (OSPE, 6), Identification (P-ID, 4), Practical performance (P-PRF, 4)	Performance directly. The only evidence that settles a doing objective.
Over time	Survey (P-SUR, 11), Compilation (COM, 6), Practical record (P-REC, 4)	Habit and consistency. The right evidence for attitude objectives.
Product	Presentation (PRN, 25), Poster (M-POS, 14), Chart (M-CHT, 2), Poster (P-POS, 1)	The result. Say in advance what it will be marked against.

Counts are how many objectives name each tool. Where a doing objective is checked only by a spoken tool, add your own observation: a checklist ticked while the student works is enough, and it is the only record showing the procedure was performed.

PART C · APPENDIX C9

Sources and further reading

The primary sources for the taxonomies used in this book, the curriculum it is built on, and a note on what was and was not altered.

THE TAXONOMIES

- Bloom, B. S. (ed.), *Taxonomy of Educational Objectives: The Classification of Educational Goals. Handbook I: Cognitive Domain.* New York: Longmans, Green, 1956. The original, and the scheme the AyUG-RB curriculum uses.
- Anderson, L. W. and Krathwohl, D. R. (eds.), *A Taxonomy for Learning, Teaching and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives.* New York: Longman, 2001. The revision: verb names, the swapped top two levels, and the knowledge axis.
- Krathwohl, D. R., Bloom, B. S. and Masia, B. B., *Taxonomy of Educational Objectives, Handbook II: Affective Domain.* New York: David McKay, 1964. The source of the five affective levels.
- Simpson, E. J., *The Classification of Educational Objectives in the Psychomotor Domain.* Washington: Gryphon House, 1972; and Dave, R. H., 'Psychomotor levels', in R. J. Armstrong (ed.), *Developing and Writing Behavioral Objectives*, 1970. The psychomotor levels used in the curriculum come from this later work rather than from Bloom's committee.
- Krathwohl, D. R., 'A Revision of Bloom's Taxonomy: An Overview', *Theory into Practice*, 41(4), 2002. A short account of what changed in 2001 and why.

THE CURRICULUM

- National Commission for Indian System of Medicine, Board of Ayurveda. *Course curriculum for Second Professional BAMS — Rasashastra evam Bhaishajyakalpana (AyUG-RB).* New Delhi: NCISM. Applicable from the 2021–22 batch, academic year 2023–24 onwards. Tables 2, 3, 4 and 4a are the source of every objective, code, priority, method and tool in this book.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words, with two exceptions, both mechanical. Plain-English spelling errors in the source have been corrected — *Discribe* to *Describe*, *Expalin* to *Explain*, and about thirty others — and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered anywhere, and no subject content has been added, removed or reworded.

The counts, level assignments and tables in Part C are generated from the same data as the Units, so the two can never disagree. Where a judgement had to be made — chiefly in reading the kind of knowledge from an objective's wording — it is made from the verb and its grammatical object, marked with a small circle (°) where the wording supports a reading without settling it, and left visible for the reader to overrule.